A Direct Fourth-Order Causal-Set Operator: Curvature Corrections and Fluctuations

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Finite-layer causal-set operators reproduce second-order differential geometry, while higher-curvature invariants have also been obtained by iterating lower-order operators. Here we construct a direct fourth-order operator and determine its Poisson-averaged continuum limit in every integer spacetime dimension $d \geq 2$. The limit is $\Box^2\phi$ supplemented by a fully determined curvature completion: couplings to $R\Box\phi$, $R_{\mu\nu}\nabla^\mu\nabla^\nu\phi$, $\nabla_\mu R\nabla^\mu\phi$, $\Box R\phi$, $R^2\phi$, $R_{\mu\nu}R^{\mu\nu}\phi$ and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\phi$. All coefficients are explicit functions of d, with the coefficient of $\Box^2$ normalized to unity. The resulting local operator has curvature couplings selected by a single finite-layer kernel, rather than inherited from the composition of a second-order operator. Its constant-field sector yields a definite general-dimensional curvature-squared density, including a $\Box R$ contribution. The sharp operator nevertheless has variance of order ℓ^{-8}, so convergence of its mean does not imply small fluctuations on individual sprinklings. We show that a mesoscopically averaged kernel preserves the same continuum limit in the mean, while its variance scales as ℓ^d/ℓ_k^{d+8} under suitable boundedness assumptions. This identifies a two-scale continuum regime in which the averaging scale vanishes while fluctuations are suppressed.

I. INTRODUCTION

A central problem in quantum gravity is to understand how the continuum description of spacetime emerges from a more fundamental microscopic structure. In the low-energy regime, general relativity admits an effective field theory description in which the Einstein–Hilbert action provides the leading contribution, while higher orders in the derivative expansion introduce corrections involving higher powers of curvature and covariant derivatives [1–3]. In four spacetime dimensions, the four-derivative sector contains the curvature invariants R^2, $R_{\mu\nu}R^{\mu\nu}$, and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, together with derivative terms such as $\Box R$. Although these terms are related by the Gauss–Bonnet identity and integrations by parts at the level of the integrated action, their coefficients and relative combinations encode important information about gravitational dynamics. They arise naturally in effective descriptions of gravity and are relevant to questions concerning ultraviolet completion and renormalization-group flow [1–3]. From a microscopic perspective, the problem is not merely to write down higher-curvature terms in a continuum action, but to identify geometric observables whose definitions do not presuppose a continuum metric. A successful microscopic construction should recover the appropriate local continuum operators while specifying how their coefficients and curvature corrections follow from the underlying structure.

Causal set theory provides a concrete framework in which to investigate this question. Its fundamental structure is a locally finite partially ordered set, with the order encoding causal relations and cardinality providing a statistical measure of spacetime volume [4, 5]. A causal set can be obtained by a Poisson sprinkling into a Lorentzian manifold, where the points are selected with density $\rho = \ell^{-d}$ and inherit the causal order of the manifold. In this setting, continuum geometric quantities can be investigated through expectation values of observables defined entirely in terms of causal order and interval cardinalities.

A particularly successful example is the causal-set d'Alembertian. Benincasa and Dowker constructed retarded finite-layer operators whose continuum limits recover the scalar d'Alembertian in flat spacetime and a curvature-corrected operator in curved spacetime [6]. The construction was extended to general spacetime dimensions by Dowker and Glaser [7]. A closed-form expression for the corresponding dimension-dependent coefficients was subsequently obtained by Glaser [8]. Belenchia, Benincasa, and Dowker established the curved-spacetime continuum limit of the four-dimensional operator under suitable assumptions on the spacetime and the scalar field [9]. Generalized causal-set d'Alembertians and their continuum properties have also been studied in a broader family of Lorentz-invariant nonlocal operators [10]. These developments show that a finite-layer causal-set prescription can reproduce nontrivial differential geometry without introducing a continuum coordinate lattice or a preferred spatial orientation. The same construction has an important action-level interpretation. Applying the causal-set curvature estimator to a constant field and summing over causal-set elements yields the Benincasa–Dowker gravitational action, whose continuum interpretation is related to the Einstein–Hilbert action [6]. The continuum behaviour of this action, including its boundary contributions, has been investigated in several works [11–13].

At the same time, causal-set d'Alembertians exhibit an important tension between nonlocality, convergence, and fluctuations. The mean of a discrete operator can approach its continuum counterpart without the fluctu-

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ations on individual sprinklings becoming negligible. A mesoscale larger than the fundamental discreteness scale has been introduced to control the large Poisson fluctuations of causal-set d'Alembertians while retaining an appropriate continuum regime [5, 14]. The associated nonlocal causal-set operators have also been investigated in the context of scalar quantum field theory [15]. Recent work has also revisited the possibility of constructing local causal-set d'Alembertians with controlled continuum behaviour [16]. These questions motivate examining higher-derivative operators at both the level of their mean continuum limit and their statistical fluctuations.

Higher-curvature operators have also been constructed by iterating lower-order causal-set operators. In particular, de Brito, Eichhorn, and Pfeiffer showed that successive applications of the causal-set d'Alembertian yield a well-defined continuum limit and, at fourth derivative order, the constant-field combination $R^2 - 2\Box R$ [17]. The $\Box R$ term is a total derivative at the level of the bulk action, in the absence of boundary contributions, so this construction provides a causal-set realization of an R^2 contribution. The curvature content of the resulting operator, however, is inherited from the composition of the same second-order operator. In particular, its fourth-order constant-field limit does not contain the additional curvature invariants $R_{\mu\nu}R^{\mu\nu}$ and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, which can enter the general local curvature-squared sector. Thus, although iteration provides a natural route to higher powers of curvature, it does not determine a general curvature-squared combination with contributions from all three quadratic curvature invariants. Obtaining such a combination requires a construction whose fourth-order curvature content is determined by the kernel itself, rather than inherited from a prescribed composition.

We address this gap by constructing a retarded finite-layer causal-set operator directly at fourth-derivative order. For each integer spacetime dimension $d \geq 2$, the operator is defined by a finite sum over past causal layers, with an overall scaling ℓ^{-4} and dimension-dependent coefficients. The layer polynomial is generated by a density-dependent Euler operator that eliminates the unwanted density powers in the large-density expansion. The remaining normalization coefficients are fixed by cancelling the leading contribution proportional to $\phi(x)$ and setting the coefficient of $\Box^2\phi(x)$ to unity. Unlike an iterated construction, this prescription determines the curvature corrections from the fourth-order kernel itself. In particular, the resulting continuum operator contains a definite combination of R^2, $R_{\mu\nu}R^{\mu\nu}$, and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, together with derivative-curvature couplings, with all coefficients explicitly determined as functions of d. The problem is therefore to establish whether this finite-layer prescription indeed yields a local covariant operator and to determine its curvature completion. The calculation requires control of the curved-spacetime Alexandrov interval volume to the corresponding order.

The geometric input to this calculation is the local expansion of the Alexandrov interval volume. Early studies established the curvature corrections to small causal-diamond volumes and clarified the role of boundary contributions [18–20]. Related investigations developed the geometry of small causal diamonds and their discrete counterparts [21]. Wang subsequently obtained a higher-order Alexandrov-volume result in the vacuum setting [22]. To the best of our knowledge, the Alexandrov-volume expansion through fourth coordinate order retaining general non-vacuum curvature contributions has not previously been presented for arbitrary spacetime dimension. We derive this expansion in Riemann normal coordinates and present the result in Appendix A. It supplies the geometric input needed to evaluate the Poisson average of the finite-layer kernel and determine its curvature-dependent continuum terms. The Alexandrov-volume expansion and the subsequent symbolic tensor calculations, including angular averaging and tensor-coefficient reduction, were performed using Cadabra [23–25]; the corresponding scripts are available in the accompanying GitHub repository .

For an integer spacetime dimension $d \geq 2$, we consider the retarded finite-layer ansatz

$$\mathcal{B}_4^{(d)}\phi(x) = \frac{1}{\ell^4}\left[\alpha_d\phi(x) + \beta_d \sum_{i=1}^{n_d} C_i^{(d)} \sum_{y\in L_i(x)} \phi(y)\right], \quad (1)$$

where $L_i(x)$ is the ith past causal layer, and $C_i^{(d)}$, α_d, and β_d are dimension-dependent coefficients. The number of layers is $n_d = \lfloor d/2 \rfloor + 4$, so that the construction has finite range for every fixed dimension. The layer coefficients are generated by a density-dependent Euler polynomial. The explicit expressions for the Euler operator, $C_i^{(d)}$, α_d, and β_d are collected in Sec. II E. The resulting Poisson-averaged continuum limit is

$$\begin{aligned}\lim_{\rho\to\infty}\left\langle \mathcal{B}_4^{(d)}\phi(x)\right\rangle = \Box^2\phi(x) &- \frac{2}{5}\Box R\phi(x) \\ &- \frac{d+2}{d+1}R\Box\phi(x) \\ &- \frac{d+2}{d+1}R_{\mu\nu}\nabla^\mu\nabla^\nu\phi(x) \\ &- \nabla_\mu R\nabla^\mu\phi(x) \\ &+ \frac{3d^2+12d+10}{12(d+1)^2}R^2\phi(x) \\ &+ \frac{(3d+4)(d+2)}{12(d+1)^2}R_{\mu\nu}R^{\mu\nu}\phi(x) \\ &- \frac{d+2}{12(d+1)}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\phi(x). \quad (2)\end{aligned}$$

This result gives the normalized local scalar operator and its curvature completion in general spacetime dimension. For a constant scalar field, the terms involving derivatives of ϕ vanish, leaving a geometric density built from curvature-squared invariants and the total-derivative term proportional to $\Box R$. Since each curvature

tensor has mass dimension two, every term in this geometric density has mass dimension four. In four spacetime dimensions, this is precisely the engineering dimension required of a local gravitational Lagrangian density, so the result selects a definite four-derivative curvature contribution with a dimensionless coefficient. In dimensions other than four, an appropriate dimensionful coupling would be required to construct an action density. This dimensional agreement in four dimensions is suggestive, but does not by itself establish the continuum limit of the corresponding causal-set action. Its relation to the conventional curvature-squared action, including the treatment of total derivatives and boundary terms, requires a separate action-level analysis.

The continuum expectation value is not the only quantity of interest. Because the causal-set operator is a random variable on an individual Poisson sprinkling, its fluctuations must also be controlled if the mean continuum limit is to describe typical realizations. We therefore analyze the variance of the sharp fourth-order operator and introduce a mesoscopic averaging prescription that separates the fundamental discreteness scale from a larger averaging scale. Under the conditions derived in Sec. III, the prescription preserves the continuum operator in the mean while suppressing the fluctuations. The analysis thus addresses both the construction of a normalized higher-derivative operator and the statistical behaviour of its discrete realization.

The paper is organized as follows. Sec. II develops the continuum-limit calculation, beginning with the geometric decomposition of the integration domain and proceeding to the density cancellations and the evaluation of the local curvature contributions. The final subsection of that section collects the explicit normalization coefficients, layer coefficients, and the resulting continuum operator. Sec. III studies the fluctuations of the sharp operator and the mesoscopic averaging prescription. Sec. IV summarizes the results and discusses their implications. Appendix A presents the Alexandrov-volume expansion in Riemann normal coordinates, while Appendix B collects the covariantization rules and auxiliary integral and coefficient identities used in the continuum calculation.

II. CONTINUUM-LIMIT ANALYSIS

Throughout the paper, we work in a d-dimensional Lorentzian spacetime with metric signature $(-,+,\dots,+)$. The connection is the Levi-Civita connection, which is metric-compatible and torsion-free; in particular, $\Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\nu\mu}$. We fix the Riemann-tensor sign convention by $[\nabla_\mu, \nabla_\nu]V_\rho = R_{\mu\nu\rho}{}^{\sigma}V_\sigma$.

With these conventions, we consider the finite-layer causal-set operator

$$\mathcal{B}_4^{(d)}\phi(x) = \frac{1}{\ell^4}\left[\alpha_d\phi(x) + \beta_d \sum_{i=1}^{n_d} C_i^{(d)} \sum_{y\in L_i(x)} \phi(y)\right], \quad (3)$$

where $L_i(x)$ denotes the ith past layer of x and n_d is the number of layers. We introduce

$$\rho := \ell^{-d}, \qquad \ell^{-4} = \rho^{4/d}, \quad (4)$$

and define the layer polynomial

$$\mathcal{P}_d(z) := \sum_{i=1}^{n_d} C_i^{(d)} \frac{z^{i-1}}{(i-1)!}. \quad (5)$$

Poisson averaging then gives

$$\left\langle \mathcal{B}_4^{(d)}\phi(x)\right\rangle = \alpha_d\rho^{4/d}\phi(x) + \beta_d\rho^{1+4/d}\int_{J^-(x)} d^dy\sqrt{-g(y)}\phi(y)\mathcal{P}_d\left(\rho V(x,y)\right)e^{-\rho V(x,y)}, \quad (6)$$

where $V(x,y)$ is the volume of the Alexandrov interval $A(x,y) = J^+(y)\cap J^-(x)$.

The density-dependent polynomial will be obtained from an Euler operator. We define

$$D_\rho := \rho\frac{\partial}{\partial\rho}. \quad (7)$$

The construction below determines the minimal polynomial whose action on $e^{-\rho V}$ removes all algebraically nonvanishing contributions from the region down the light cone, thereby generating the layer factor $P_d(\rho V)e^{-\rho V}$.

A. Geometric Setup and Decomposition

Let

$$K := \operatorname{supp}\phi \quad (8)$$

denote the compact support of the test field. Thus only $K\cap J^-(x)$ contributes to Eq. (6). We take x to be the origin of a Riemann-normal-coordinate (RNC) neighbourhood $\mathcal{N}$ and introduce the past-directed radial null coordinates

$$u = \frac{-t-r}{\sqrt{2}}, \qquad v = \frac{-t+r}{\sqrt{2}}, \quad (9)$$

so that locally the past light cone of x is $u = 0$ and the causal past is described by $0 \le u \le v$. The RNC

description is convenient near x, but it is not adapted to the part of the light cone extending away from the origin. We therefore introduce the portion of the past light cone relevant to the support of the field,

$$\mathcal{L} := \partial J^-(x) \cap K, \tag{10}$$

and assume that each point of $\mathcal{L}$ is connected to x by a unique past-directed null geodesic. The corresponding null generators are denoted by γ_Ω, where

$$\Omega = (\Omega^1, \ldots, \Omega^{d-2}) \tag{11}$$

labels the generators. In a neighbourhood $\mathcal{N}_{\rm LC}$ of $\mathcal{L}$ we use Null Gaussian Normal Coordinates (NGNC),

$$(U, V, \Omega), \tag{12}$$

where V is an affine parameter along a null generator and U is an affine parameter along the transverse null geodesics. The light cone itself is

$$U = 0. \tag{13}$$

The NGNC therefore provide a single coordinate system adapted to the whole null hypersurface. For the local volume estimate around any fixed generator γ_Ω, we further assume the existence of a tubular neighbourhood $\mathcal{T}_\Omega$ of that generator on which Null Fermi Normal Coordinates (NFNC) are valid. These coordinates are adapted to the individual null geodesic and its parallel-transported null frame. We choose the transverse thickness of the light-cone region sufficiently small that the complete Alexandrov interval between x and every point of W_2 remains inside the corresponding $\mathcal{T}_\Omega$. This is the geometric assumption that permits the use of the NFNC expansion and its Penrose-limit scaling. The terminology and construction follow the null Gaussian and null Fermi coordinate treatment used for the curved-spacetime light-cone analysis of Ref. [9], based on the construction of null Fermi coordinates in Ref. [26]. There is no need to force the RNC and NGNC boundaries to coincide. Indeed, they are adapted to different coordinate constructions. In curved spacetime, one generally has

$$u \neq U, \qquad v \neq V \tag{14}$$

away from the common null hypersurface. The coordinate transition map is nevertheless smooth in the overlap of the two neighbourhoods, and the two descriptions become arbitrarily close as the RNC neighbourhood is shrunk towards x. We therefore choose two small positive scales satisfying

$$0 < a' \leq a, \tag{15}$$

where a controls the RNC near region and a' is the corresponding lower affine cutoff used in NGNC. The inequality $a' \leq a$ is chosen deliberately: it allows the two regions to overlap rather than forcing an artificial separation between two different coordinate systems. This is the same geometric issue exhibited explicitly in Fig. 2 of Ref. [9], where the RNC and NGNC/NFNC boundaries are shown to be misaligned in curved spacetime.

With these coordinate setups, the near region is

$$W_1 := \{ y \in \mathcal{N} \mid 0 \leq u(y) \leq v(y) \leq a \}. \tag{16}$$

The light-cone region is defined in NGNC by

$$W_2 := \left\{ y \in \mathcal{N}_{\rm LC} \cap K \cap J^-(x) \;\middle|\; a' \leq V(y) \leq L, \quad 0 \leq U(y) \leq \frac{b^2}{[f_0(V(y), \Omega)]^{2/d}} \right\}, \tag{17}$$

where L is chosen large enough to contain the relevant support of ϕ along the light cone. The positive constant b has dimensions of length. The f_0-dependent form of the transverse cutoff is useful because it makes the leading Alexandrov volume at the upper boundary of W_2 independent of V and Ω. Indeed, once the light-cone expansion derived below is introduced,

$$V_0 = U^{d/2} f_0(V, \Omega), \tag{18}$$

and on the upper boundary of W_2, $V_0 \leq b^d$. Finally, define the remaining region by

$$W_3 := (K \cap J^-(x)) \setminus (W_1 \cup W_2). \tag{19}$$

Because W_1 and W_2 are defined in different coordinate descriptions, and because their boundaries need not coincide in curved spacetime, we do not require a disjoint partition. Instead, the overlap

$$W_{12} := W_1 \cap W_2 \tag{20}$$

is retained. If

$$\mathscr{I}[W] := \widehat{\mathcal{E}}_d \int_W d^d y \sqrt{-g(y)} \phi(y) e^{-\rho V(x,y)}, \tag{21}$$

then

$$\mathscr{I}[W_1] + \mathscr{I}[W_2] + \mathscr{I}[W_3] = \mathscr{I}[K \cap J^-(x)] + \mathscr{I}[W_{12}]. \tag{22}$$

Since

$$W_{12} \subset W_2, \tag{23}$$

the additional term is included in the light-cone analysis. Once the W_2 contribution is shown to vanish in the

continuum limit, the overlap therefore has no effect on the final result. This is the reason it is preferable here not to force a disjoint partition merely for geometrical convenience.

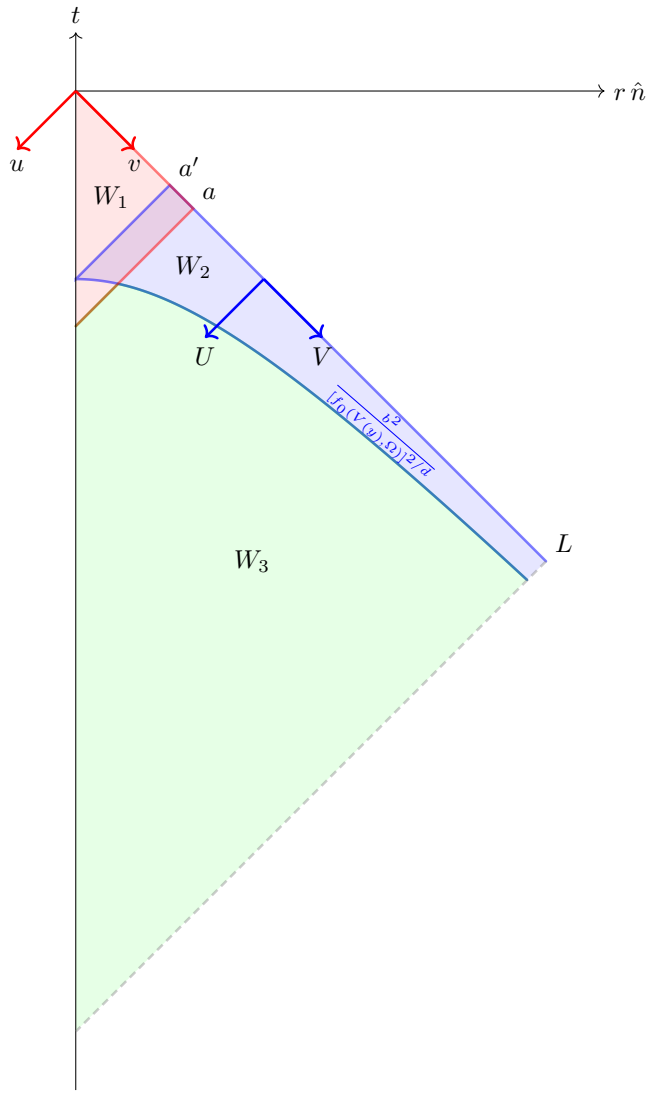


FIG. 1. Schematic geometry of the continuum-limit decomposition. The RNC-defined near region W_1 and the NGNC-defined light-cone region W_2 need not have coincident boundaries in curved spacetime, and we therefore retain their overlap. The transverse thickness of W_2 is chosen using the leading light-cone coefficient $f_0(V,\Omega)$ so that the leading Alexandrov volume remains uniformly bounded throughout the tubular neighbourhood.

B. The Deep-Past Region W_3

By the decomposition in Eqs. (16)–(19), the region W_3 is bounded away from the past light cone. The Alexandrov volume is continuous in the relevant chronological region and vanishes on the null boundary. Since W_3 is compact and does not intersect that boundary, there exists a finite strictly positive lower bound

$$V(x,y) \geq V_{\min} > 0, \qquad y \in W_3, \tag{24}$$

where

$$V_{\min} := \min_{y \in W_3} V(x,y). \tag{25}$$

The precise location of the minimum is immaterial for the continuum limit. In the flat-space construction, the corresponding lower bound can be identified explicitly from the boundary separating the near and deep regions; in particular, the flat geometry provides the familiar reference scale associated with the surface $u = v = a$. In the curved geometry, the boundary is displaced and distorted, so we do not need to identify its minimizing point. What survives from the flat argument is the invariant statement (24): once a finite neighbourhood of the light cone has been removed, the remaining compact set has a non-zero Alexandrov-volume gap.

For any finite-order density differential operator, differentiating the factor $e^{-\rho V}$ can generate only finite polynomial powers of ρ and factors involving V. Since V is bounded below on W_3, there is therefore a constant C and a finite integer N such that

$$|\mathscr{I}[W_3]| \leq C\rho^N e^{-\rho V_{\min}}. \tag{26}$$

Hence

$$\lim_{\rho\to\infty} \rho^N e^{-\rho V_{\min}} = 0, \tag{27}$$

and consequently

$$\lim_{\rho\to\infty} \mathscr{I}[W_3] = 0. \tag{28}$$

Thus, no local curvature expansion is required in W_3.

C. The Light-Cone Region W_2

Unlike the deep-past region W_3 discussed in Sec. II B, the region W_2 requires a different analysis because the Alexandrov volume approaches zero as $U \to 0$. The relevant geometry is controlled by the tubular neighbourhood of the null generators introduced above. For a fixed generator γ_Ω, introduce Null Fermi Normal Coordinates

$$(x^+, x^-, x^a), \qquad a = 1, \ldots, d-2, \tag{29}$$

with the reference generator represented by

$$x^- = 0, \qquad x^a = 0. \tag{30}$$

The endpoint y lying on the transverse null geodesic from γ_Ω is represented by

$$(x^+, x^-, x^a) = (V, U, 0, \ldots, 0). \tag{31}$$

The relevant Penrose scaling is

$$x^+ = z^+, \qquad x^- = Uz^-, \qquad x^a = \sqrt{U} z^a. \tag{32}$$

The corresponding pulled-back metric has the form

$$ds^2 = U\left[-2dz^+dz^- + \delta_{ab}dz^a dz^b - R_{+a+b}(z^+)z^a z^b (dz^+)^2 + O(U^{1/2})\right], \tag{33}$$

where the curvature components are evaluated along the reference null generator. Thus, the transverse scaling produces a Jacobian

$$U \times U^{(d-2)/2} = U^{d/2}, \tag{34}$$

and the integrated Alexandrov volume has the leading homogeneity

$$V(x,y) = U^{d/2}\left[f_0(V,\Omega) + o(1)\right], \qquad U \to 0^+. \tag{35}$$

We now define

$$f_0(V,\Omega) := \lim_{U\to 0^+} U^{-d/2}V(x,y). \tag{36}$$

The function f_0 is finite and positive for $V > 0$. It is determined by the volume of the limiting Alexandrov interval in the corresponding Penrose-limit geometry. Moreover, causal inclusion of the limiting intervals implies

$$0 < V < V' \quad \Longrightarrow \quad f_0(V,\Omega) \le f_0(V',\Omega), \tag{37}$$

so f_0 is non-decreasing along each null generator. For the asymptotic analysis, it is convenient to retain one controlled subleading order and write

$$V(x,y) = U^{d/2}f_0(V,\Omega) + U^{d/2+1}G(U,V,\Omega), \tag{38}$$

where G is bounded on the compact W_2 region. This is the general-dimensional form of the regular light-cone volume expansion. The detailed coefficients contained in G will not be required. It is useful to separate

$$V_0 := U^{d/2}f_0(V,\Omega), \qquad \delta V := U^{d/2+1}G(U,V,\Omega), \tag{39}$$

so that

$$V = V_0 + \delta V. \tag{40}$$

On the region W_2, the choice of the transverse cutoff in Eq. (17) gives

$$\frac{\delta V}{V_0} = \frac{UG}{f_0} \tag{41}$$

and hence

$$\left|\frac{\delta V}{V_0}\right| \le b^2 \frac{\sup_{W_2}|G|}{\inf_{W_2} f_0}\left[\inf_{W_2} f_0\right]^{-2/d}. \tag{42}$$

Because f_0 is positive and continuous on the compact (V,Ω) domain $[a',L]\times S^{d-2}$, the denominator is strictly positive. The parameter b can therefore be chosen sufficiently small that

$$|\delta V| \ll V_0 \tag{43}$$

uniformly throughout W_2.

The characteristic transverse scale follows from

$$\rho V_0 \sim 1, \tag{44}$$

which gives

$$U \sim \rho^{-2/d}. \tag{45}$$

On this scale,

$$\rho\delta V = \rho U^{d/2+1}G = O(\rho^{-2/d}), \tag{46}$$

so that the correction to the exponential is subleading:

$$e^{-\rho V} = e^{-\rho V_0}\left[1 + O(\rho^{-2/d})\right]. \tag{47}$$

Thus the leading density power counting is completely controlled by $U^{d/2}f_0$. To see why the f_0-dependent upper boundary in Eq. (17) is particularly convenient, consider the basic transverse integral

$$J_m(\rho) := \int_0^{b^2/f_0^{2/d}} dU U^m e^{-\rho f_0 U^{d/2}}. \tag{48}$$

With

$$s = \rho f_0 U^{d/2}, \tag{49}$$

we obtain

$$U = \left(\frac{s}{\rho f_0}\right)^{2/d}, \qquad dU = \frac{2}{d}(\rho f_0)^{-2/d}s^{2/d-1}ds, \tag{50}$$

and therefore

$$J_m(\rho) = \frac{2}{d}(\rho f_0)^{-2(m+1)/d}\gamma\left(\frac{2(m+1)}{d}, \rho b^d\right), \tag{51}$$

where γ is the lower incomplete Gamma function. Notice that the upper argument of the incomplete Gamma function is independent of V and Ω. Consequently, for fixed $b > 0$,

$$J_m(\rho) = \frac{2}{d}\Gamma\left(\frac{2(m+1)}{d}\right) f_0(V,\Omega)^{-2(m+1)/d}\rho^{-2(m+1)/d} + \mathcal{O}\left(e^{-\rho b^d}\right). \tag{52}$$

The measure and the scalar field are smooth in the transverse coordinate, so their product may be expanded locally as

$$\sqrt{-g(y)}\phi(y)d^dy = \left[\sum_{m=0}^{M} A_m(V,\Omega)U^m + O(U^{M+1})\right] \times dU dV d\Omega_{d-2}, \tag{53}$$

for any finite M required in the asymptotic argument. Combining this expansion with Eq. (52), the contribution of a term of transverse order U^m to the integral appearing before the Euler operator is

$$\rho^{-2(m+1)/d}. \tag{54}$$

The overall layer contribution in Eq. (6) carries the factor $\rho^{1+4/d}$. Thus, before cancellation, a term of order U^m contributes with the density power

$$\rho^{1+4/d-2(m+1)/d} = \rho^{(d+2-2m)/d}. \tag{55}$$

All terms which do not vanish automatically in the light-cone asymptotic expansion must be annihilated by the density operator. Equivalently, the powers occurring in the transverse integral that need to be removed are

$$\rho^{-2j/d}, \qquad j = 1, \dots, q_d, \tag{56}$$

with

$$q_d = \left\lfloor \frac{d+2}{2} \right\rfloor + 1. \tag{57}$$

Since

$$D_\rho \rho^\lambda = \lambda \rho^\lambda, \tag{58}$$

the minimal Euler polynomial required by the light-cone region alone is

$$\widehat{\mathcal{E}}_d^{\mathrm{LC}} = \mathcal{N}_d^{\mathrm{LC}} \prod_{j=1}^{q_d} \left(D_\rho + \frac{2j}{d} \right), \tag{59}$$

where

$$\mathcal{N}_d^{\mathrm{LC}} = \left[\prod_{j=1}^{q_d} \frac{2j}{d} \right]^{-1} = \frac{d^{q_d}}{2^{q_d} q_d!}. \tag{60}$$

This operator is fixed by the W_2 power counting. The near-region analysis below imposes one additional condition, so $\widehat{\mathcal{E}}_d^{\mathrm{LC}}$ is the light-cone factor of the final density operator rather than the final operator itself. Define

$$\mathscr{I}[W_2] := \widehat{\mathcal{E}}_d^{\mathrm{LC}} \int_{W_2} d^d y \sqrt{-g(y)} \phi(y) e^{-\rho V(x,y)}. \tag{61}$$

The preceding power counting gives

$$\mathscr{I}[W_2] = O\left(\rho^{-2(q_d+1)/d} \right), \tag{62}$$

and hence the W_2 contribution vanishes in the continuum limit. The final operator constructed below contains $\widehat{\mathcal{E}}_d^{\mathrm{LC}}$ as a factor, so this conclusion is unchanged by the subsequent W_1 refinement.

D. The Near Region W_1

The near region W_1 contains the point x and is therefore the only region that can contribute a genuinely local term in the continuum limit. The region W_1, defined in Eq. (16), is the neighbourhood of $x = 0$ described by the RNC coordinates. Using the null coordinates introduced in Eq. (9), the coordinate volume element becomes

$$d^d y = 2^{-(d-2)/2} (v-u)^{d-2} du dv d\Omega_{d-2}. \tag{63}$$

We define

$$I_1(\rho) := \widehat{\mathcal{E}}_d \int_{W_1} d^d y \sqrt{-g(y)} \phi(y) e^{-\rho V(x,y)}. \tag{64}$$

The corresponding contribution to the expectation value in Eq. (6) is

$$\mathcal{B}^{(d)}_{4,W_1} \phi(x) = \beta_d \rho^{1+4/d} I_1(\rho). \tag{65}$$

a. Local Expansions and Truncation. The relevant large-density behaviour in W_1 is governed by the flat Alexandrov volume, which is the leading term in the general expansion derived in Eq. (A52). In particular, from Eq. (A51),

$$V_0(y) := V^{(d)}(y) = \frac{\Omega_{d-2}}{2^{d-1} d(d-1)} \tau^d, \qquad \tau^2 = 2uv. \tag{66}$$

Thus, ρV_0 is of order unity in the region that controls the large-ρ asymptotics. We assign the near-region weights

$$\mathrm{ord}_{\mathrm{nr}}(y^\mu) = 1, \qquad \mathrm{ord}_{\mathrm{nr}}(\rho) = -d, \qquad \mathrm{ord}_{\mathrm{nr}}(\rho V_0) = 0. \tag{67}$$

Since V_0 has coordinate order d, a relative correction of coordinate order k to the Alexandrov volume produces

$$\mathrm{ord}_{\mathrm{nr}} \left(\rho \delta V_{(k)} \right) = k. \tag{68}$$

Because the overall layer contribution in Eq. (65) carries $\rho^{1+4/d}$, terms of effective order four are the last terms that can contribute to a finite continuum limit. We therefore retain all terms through effective order four.

The scalar field is expanded through fourth order,

$$\phi(y) = \phi + \phi_{,\mu} y^\mu + \frac{1}{2} \phi_{,\mu\nu} y^\mu y^\nu + \frac{1}{3!} \phi_{,\mu\nu\rho} y^\mu y^\nu y^\rho + \frac{1}{4!} \phi_{,\mu\nu\rho\sigma} y^\mu y^\nu y^\rho y^\sigma + \mathcal{R}^{(5)}_\phi(y), \tag{69}$$

where all derivatives are evaluated at $x = 0$, and

$$\mathcal{R}^{(5)}_\phi(y) = y^\mu y^\nu y^\rho y^\sigma y^\lambda \Psi_{\mu\nu\rho\sigma\lambda}(y). \tag{70}$$

The function $\Psi_{\mu\nu\rho\sigma\lambda}$ is bounded on the compact region W_1. For the invariant volume density, we use the RNC expansion derived in Eq. (A7) of Appendix A,

$$\sqrt{-g(y)} = M_{\le 4}(y) + \mathcal{R}^{(5)}_g(y), \tag{71}$$

where

$$M_{\le 4}(y) = 1 + M_{(2)}(y) + M_{(3)}(y) + M_{(4)}(y) \tag{72}$$

contains the explicitly displayed terms in Eq. (A7), while

$$\mathcal{R}_g^{(5)}(y) = y^\mu y^\nu y^\rho y^\sigma y^\lambda T_{\mu\nu\rho\sigma\lambda}(y), \tag{73}$$

with $T_{\mu\nu\rho\sigma\lambda}$ bounded on W_1. For the Alexandrov volume, we use the compact expansion in Eq. (A53) of Appendix A,

$$V(x,y) = V_0(y)\left[1 + \mathcal{V}_{(2)}(y) + \mathcal{V}_{(4)}(y) + \mathcal{O}(y^5)\right]. \tag{74}$$

Equivalently, defining

$$\delta V_{(k)}(y) := V_0(y)\mathcal{V}_{(k)}(y), \qquad k = 2, 4, \tag{75}$$

we have

$$V(x,y) = V_0(y) + \delta V_{(2)}(y) + \delta V_{(4)}(y) + \mathcal{R}_V^{(5)}(y), \tag{76}$$

where

$$\mathcal{R}_V^{(5)}(y) = V_0(y) O_{\rm nr}(5). \tag{77}$$

The explicit expressions for $\mathcal{V}_{(2)}$, $\mathcal{V}_{(3)}$ and $\mathcal{V}_{(4)}$ are given in Eqs. (A54)-(A55).

It is convenient to separate the flat exponential,

$$e^{-\rho V(x,y)} = e^{-\rho V_0(y)} e^{-\rho \delta V(y)}, \qquad \delta V := V - V_0. \tag{78}$$

Through effective order four,

$$e^{-\rho\delta V} = 1 - \rho\delta V_{(2)} - \rho\delta V_{(4)} + \frac{1}{2}\rho^2 \delta V_{(2)}^2 + O_{\rm nr}(5). \tag{79}$$

The quadratic term $\frac{1}{2}\rho^2\delta V_{(2)}^2$ must be retained because it is itself of effective order four. The first omitted terms have effective order five and are included in the remainder $O_{\rm nr}(5)$.

b. Decomposition of the Integrand. Define

$$\phi_{\le 4}(y) = \phi + \phi_{,\mu} y^\mu + \frac{1}{2}\phi_{,\mu\nu} y^\mu y^\nu + \frac{1}{3!}\phi_{,\mu\nu\rho} y^\mu y^\nu y^\rho + \frac{1}{4!}\phi_{,\mu\nu\rho\sigma} y^\mu y^\nu y^\rho y^\sigma, \tag{80}$$

and

$$\mathscr{E}_{\le 4} = 1 - \rho\delta V_{(2)} - \rho\delta V_{(4)} + \frac{1}{2}\rho^2\delta V_{(2)}^2. \tag{81}$$

The integrand may then be organized as

$$\sqrt{-g(y)}\phi(y) e^{-\rho V(x,y)} = \left[A(y) + B(y) + C(y)\right] e^{-\rho V_0(y)}, \tag{82}$$

where A contains all explicitly known local terms through effective order four,

$$A(y) := \left[M_{\le 4}(y)\phi_{\le 4}(y)\mathscr{E}_{\le 4}(y)\right]_{\le 4}, \tag{83}$$

with $[\cdots]_{\le 4}$ denoting truncation to effective order at most four. The group B contains every term involving at least one of the remainders

$$\mathcal{R}_\phi^{(5)}, \qquad \mathcal{R}_g^{(5)}, \qquad \mathcal{R}_V^{(5)}. \tag{84}$$

Finally, C contains the terms generated by the part of the exponential expansion beyond effective order four. Thus the three groups are distinguished by their mathematical role: A is the explicit local sector to be evaluated, B contains the Taylor and volume remainders, and C is the exponential remainder.

c. Vanishing of B. Every contribution to B contains at least five effective powers of the near-region coordinates. To estimate its large-ρ behaviour, consider a generic monomial of coordinate degree $m + n$. After expansion of the factor $(v-u)^{d-2}$, its contribution is reduced to the basic double-integral family of Appendix B 2. The relevant asymptotic sector scales as

$$\rho^{-(m+n+2)/d}. \tag{85}$$

For an effective-order-s term,

$$m + n = d - 2 + s, \tag{86}$$

so that, after restoring the overall factor $\rho^{1+4/d}$,

$$\rho^{1+4/d}\rho^{-(m+n+2)/d} = \rho^{(4-s)/d}. \tag{87}$$

Since $s \ge 5$ for B, this contribution is suppressed by at least one negative power of ρ. The alternative asymptotic sector gives powers $\rho^{-2(m+1)/d}$. Whenever $m + 1 \le q_d$, these are annihilated by the Euler operator $\widehat{\mathcal{E}}_d$ defined in Eq. (102). For $m + 1 > q_d$, the resulting density power is already negative after the overall factor in Eq. (65) is restored. Possible logarithmic factors in the diagonal case do not alter the vanishing conclusion. Therefore,

$$\rho^{1+4/d} I_B(\rho) \longrightarrow 0, \qquad \rho \to \infty. \tag{88}$$

d. Vanishing of C. The group C contains the terms generated by the part of the exponential expansion beyond effective order four. Since the retained volume corrections have effective orders two and four, the first omitted products generated by their exponential expansion are $\rho^2\delta V_{(2)}\delta V_{(4)}$ and $\rho^3\delta V_{(2)}^3$, both of effective order six. Consequently,

$$\mathrm{ord}_{\rm nr}\, C \ge 6. \tag{89}$$

The same asymptotic estimate used for B then gives

$$\rho^{1+4/d} I_C(\rho) \longrightarrow 0, \qquad \rho \longrightarrow \infty. \tag{90}$$

Thus the continuum-limit calculation in the near region reduces to the explicit local sector A alone.

e. Asymptotic Sector Classification. The explicit local sector in Eq. (83) is most efficiently analyzed by separating the effective coordinate order from the powers of the zeroth-order exponential variable. Write

$$A(y) = \sum_{p=0}^{4} A_{(p)}(u,v), \tag{91}$$

where p denotes the total explicit coordinate order. With

$$Z_0 := \rho V_0 = \kappa_d \rho (uv)^{d/2}, \qquad \kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1} d(d-1)}, \tag{92}$$

a typical term in a fixed Z_0^q sector may be written as

$$Z_0^q u^c v^e, \qquad c+e=p. \tag{93}$$

After multiplication by the radial measure $(v-u)^{d-2}$, one obtains monomials

$$Z_0^q (v-u)^{d-2} u^c v^e \propto \rho^q \sum_{k=0}^{d-2} (-1)^k \binom{d-2}{k} u^M v^N, \tag{94}$$

where

$$M = k + c + \frac{qd}{2}, \qquad N = d-2-k+e+\frac{qd}{2}. \tag{95}$$

Hence

$$M-N = 2(k+c)-(d-2+p), \qquad M+N = d-2+p+qd. \tag{96}$$

In particular, the condition $M=N$ is independent of the Z_0 sector.

Using the asymptotic decomposition of the double integrals in Appendix B 2, the two sectors scale as

$$Z_0^q \mathcal{I}^{(1)}_{M,N} \sim \rho^{-2(k+c+1)/d}, \tag{97}$$

$$Z_0^q \mathcal{I}^{(2)}_{M,N} \sim \rho^{-(d+p)/d}. \tag{98}$$

The first sector is already controlled by the light-cone Euler roots: all non-decaying powers are among $\rho^{-2j/d}$ with $j \le q_d$, while terms with larger j are already suppressed after the overall factor $\rho^{1+4/d}$ is restored. It remains to classify the second asymptotic sector. After angular averaging, the coefficient of each fixed (p,q) sector is symmetric under

$$A_{(p),q}(u,v) = A_{(p),q}(v,u). \tag{99}$$

The measure factor has parity

$$(v-u)^{d-2} \longrightarrow (-1)^{d-2}(v-u)^{d-2} \qquad (u \leftrightarrow v). \tag{100}$$

For even d, the full integrand is symmetric. Since the second asymptotic sector satisfies

$$\mathcal{I}^{(2)}_{M,N} = -\mathcal{I}^{(2)}_{N,M}, \qquad M \neq N. \tag{101}$$

All off-diagonal contributions cancel pairwise. For odd d, the full integrand is antisymmetric. Consequently, all diagonal coefficients vanish, while the off-diagonal terms remain subject only to the density scaling in Eq. (98). The resulting classification through $p=4$ is summarized in Table I.

f. Final Density Operator. The W_1 classification shows that one additional Euler factor is required to eliminate the $p=2$ sector. The final density operator is therefore

$$\widehat{\mathcal{E}}_d = \mathcal{N}_d \left(D_\rho + \frac{d+2}{d}\right) \prod_{j=1}^{q_d} \left(D_\rho + \frac{2j}{d}\right). \tag{102}$$

The normalization for the unit constant term is

$$\mathcal{N}_d = \frac{d^{q_d+1}}{2^{q_d} q_d! (d+2)}. \tag{103}$$

The role of the additional factor depends on the parity of d. For odd d, the $p=2$ power

$$\rho^{-(d+2)/d} \tag{104}$$

is not among the light-cone roots, so the new factor supplies the missing simple root. For even d, the same root is already present once in the light-cone operator, but the diagonal integral has the logarithmic form

$$\rho^{-(d+2)/d} \left(A \ln \rho + B\right). \tag{105}$$

The additional factor therefore makes this root double and removes both the logarithmic and non-logarithmic pieces. The extra factor does not remove the sectors required for the continuum limit. For

$$\lambda_0 = -1, \qquad \lambda_4 = -\frac{d+4}{d}, \tag{106}$$

one has

$$\lambda_0 + \frac{d+2}{d} = \frac{2}{d} \neq 0, \qquad \lambda_4 + \frac{d+2}{d} = -\frac{2}{d} \neq 0. \tag{107}$$

Thus the additional factor removes $p=2$ without removing either the leading $p=0$ contribution or the finite $p=4$ contribution.

Since the final operator has degree q_d+1 in D_ρ, the number of layers is

$$n_d = q_d + 2 = \left\lfloor \frac{d+2}{2} \right\rfloor + 3. \tag{108}$$

The corresponding layer polynomial is defined by

$$\widehat{\mathcal{E}}_d e^{-z} = \mathcal{P}_d(z) e^{-z}, \qquad z = \rho V, \tag{109}$$

with

$$\mathcal{P}_d(z) = \sum_{i=1}^{n_d} C_i^{(d)} \frac{z^{i-1}}{(i-1)!}. \tag{110}$$

g. Sectorwise Evaluation of the Local Sector. The explicit local contribution is organized as

$$A(y) = \sum_{p=0}^{4} A_{(p)}(u,v). \tag{111}$$

	d even		d odd	
p	$M \neq N$	$M = N$	$M \neq N$	$M = N$
0	pair cancellation	leading term, fixes α_d	leading term, fixes α_d	no diagonal
1	pair cancellation	no diagonal	old Euler root	antisymmetry $\Rightarrow 0$
2	pair cancellation	diagonal/log sector	missing Euler root	no diagonal
3	pair cancellation	no diagonal	old Euler root	antisymmetry $\Rightarrow 0$
4	pair cancellation	finite local term	finite off-diagonal term	no diagonal

TABLE I. Classification of the second asymptotic sector in W_1. For even d, off-diagonal terms cancel pairwise, while for odd d diagonal coefficients vanish by antisymmetry of the measure. The $p = 1$ and $p = 3$ sectors are otherwise already removed by the light-cone Euler roots. The only additional unwanted sector is $p = 2$: it is diagonal and logarithmic for even d, and off-diagonal for odd d. The $p = 0$ sector supplies the leading normalization term, while $p = 4$ supplies the finite local contribution.

As established above, the sectors $p = 1, 2, 3$ do not contribute in the continuum limit. We therefore retain $p = 0, 4$. With

$$d^d y = 2^{-(d-2)/2}(v-u)^{d-2} du dv d\Omega_{d-2}, \tag{112}$$

and

$$Z_0 = \sigma_d (uv)^{d/2}, \qquad \sigma_d = \kappa_d \rho, \qquad \kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1} d(d-1)}, \tag{113}$$

we use the normalized angular average

$$\langle F \rangle_\Omega := \frac{1}{\Omega_{d-2}} \int_{S^{d-2}} d\Omega_{d-2} F. \tag{114}$$

The quantities $A_{(p),q}$ and $K_m^{(p,q)}$ below are understood as angular averages in this sense. The angular measure is restored through

$$\int_{S^{d-2}} d\Omega_{d-2} F = \Omega_{d-2} \langle F \rangle_\Omega. \tag{115}$$

For fixed p and $q = 0, 1, 2$, write

$$A_{(p),q}(u,v) = \sum_{m=0}^{p} K_m^{(p,q)} u^{m+\frac{qd}{2}} v^{p-m+\frac{qd}{2}}. \tag{116}$$

Multiplying by the radial measure gives

$$(v-u)^{d-2} A_{(p),q}(u,v) = \sum_{r=0}^{d-2+p} L_r^{(p,q)} u^{r+\frac{qd}{2}} v^{d-2+p-r+\frac{qd}{2}}, \tag{117}$$

where

$$L_r^{(p,q)} = \sum_{m=0}^{p} (-1)^{r-m} \binom{d-2}{r-m} K_m^{(p,q)}, \tag{118}$$

with the binomial coefficient understood to vanish for $r - m \notin \{0, \dots, d-2\}$.

Using the double-integral identity of Appendix B 2, the non-diagonal terms separate into the two asymptotic sectors

$$\mathcal{I}_r^{(p,q,1)} \sim \frac{2L_r^{(p,q)} a^{d-2+p-2r}}{d(d-2+p-2r)} \Gamma\left(q + \frac{2(r+1)}{d}\right) \sigma_d^{-2(r+1)/d}, \tag{119}$$

$$\mathcal{I}_r^{(p,q,2)} \sim -\frac{2L_r^{(p,q)}}{d(d-2+p-2r)} \Gamma\left(q + \frac{d+p}{d}\right) \sigma_d^{-(d+p)/d}, \tag{120}$$

for $r \neq r_\star^{(p)}$, where

$$r_\star^{(p)} = \frac{d-2+p}{2}. \tag{121}$$

The first sector scales as

$$\rho^{1+4/d} \mathcal{I}_r^{(p,q,1)} \sim \rho^{(d+2-2r)/d}. \tag{122}$$

Whenever this power is non-decaying, the exponent $-2(r+1)/d$ is one of the Euler roots of $\widehat{\mathcal{E}}_d$; the remaining values of r are already suppressed. Thus the first sector is removed for both even and odd d. The second sector scales as

$$\rho^{1+4/d} \mathcal{I}_r^{(p,q,2)} \sim \rho^{(4-p)/d}. \tag{123}$$

Thus $p = 0$ produces the leading $\rho^{4/d}$ scaling, while $p = 4$ produces a finite scaling. For even d, the off-diagonal terms cancel pairwise, leaving only the diagonal logarithmic sector; for odd d, there is no diagonal term for $p = 0, 4$, and the off-diagonal second sector remains. The diagonal condition is

$$M = N \quad \Longleftrightarrow \quad r = r_\star^{(p)} = \frac{d-2+p}{2}, \tag{124}$$

which is possible for $p = 0, 4$ only when d is even. Hence the logarithmic sector is absent for odd d. For even d, the diagonal contribution has the asymptotic form

$$\mathcal{I}_{r_\star}^{(p,q)} \sim \frac{2L_{r_\star}^{(p,q)}}{d^2} \Gamma\left(q + \frac{d+p}{d}\right) \sigma_d^{-(d+p)/d} \times \left[\ln(\sigma_d a^d) - \psi\left(q + \frac{d+p}{d}\right)\right]. \tag{125}$$

Thus the logarithmic sector scales as

$$\mathcal{I}^{(p,q)}_{r_\star} \sim \rho^{-(d+p)/d} \ln \rho. \tag{126}$$

Since $\widehat{\mathcal{E}}_d \rho^\lambda = Q_d(\lambda)\rho^\lambda$ and $Q_d(-(d+p)/d) = 0$ for $p = 0, 4$ and even d,

$$\widehat{\mathcal{E}}_d \left[\rho^{-(d+p)/d} \ln \rho\right] = Q'_d\left(-\frac{d+p}{d}\right) \rho^{-(d+p)/d}. \tag{127}$$

Consequently, the surviving structure is entirely fixed by the parity of d: odd d gives the second asymptotic sector, while even d gives the logarithmic sector.

Odd-dimensional case. For odd d and $p = 0, 4$, the integer $d-2+p$ is odd, so there is no diagonal term. The surviving terms occur in pairs:

$$r' = d - 2 + p - r \qquad 0 \le r \le \frac{d-3+p}{2}. \tag{128}$$

Since the angular-averaged polynomial is symmetric under $u \leftrightarrow v$,

$$L^{(p,q)}_{r'} = (-1)^{d-2} L^{(p,q)}_r = -L^{(p,q)}_r, \tag{129}$$

and therefore

$$\mathcal{I}^{(p,q,2)}_r + \mathcal{I}^{(p,q,2)}_{r'} = -\frac{4L^{(p,q)}_r \sigma_d^{-(d+p)/d}}{d(d-2+p-2r)} \Gamma\left(q + \frac{d+p}{d}\right). \tag{130}$$

Substituting the definition of $L^{(p,q)}_r$ and writing $r = m+j$ gives

$$\sum_{r=0}^{d-2+p} \frac{L^{(p,q)}_r}{d-2+p-2r} = \sum_{m=0}^{p} \sum_{j=0}^{d-2} \frac{(-1)^j \binom{d-2}{j} K^{(p,q)}_m}{d-2+p-2m-2j}. \tag{131}$$

The inner binomial sum is evaluated in closed form using

$$\sum_{j=0}^{n} \frac{(-1)^j \binom{n}{j}}{a-j} = (-1)^n \frac{\Gamma(n+1)\Gamma(a-n)}{\Gamma(a+1)}. \tag{132}$$

Setting $n = d-2$ and $a = (d-2+p-2m)/2$ gives

$$\sum_{r=0}^{(d-3+p)/2} \frac{L^{(p,q)}_r}{d-2+p-2r} = \sum_{m=0}^{p} \frac{-\Gamma\left(\frac{p+2-d-2m}{2}\right) K^{(p,q)}_m}{4(\Gamma(d-1))^{-1}\Gamma\left(\frac{d+p-2m}{2}\right)}. \tag{133}$$

Hence, the complete odd-dimensional contribution from the near region is

$$I_1^{\mathrm{odd}}(\rho) \sim \frac{\Omega_{d-2}\Gamma(d-1)}{2^{(d-2)/2} d} \sum_{p\in\{0,4\}} \sum_{q=0}^{2} \Gamma\left(q + \frac{d+p}{d}\right) Q_d\left(-\frac{d+p}{d}\right) \sigma_d^{-(d+p)/d} \sum_{m=0}^{p} K^{(p,q)}_m \frac{\Gamma\left(\frac{p+2-d-2m}{2}\right)}{\Gamma\left(\frac{d+p-2m}{2}\right)}. \tag{134}$$

For odd d, $Q_d(-1) \neq 0$ and $Q_d(-(d+4)/d) \neq 0$, so the $p = 0$ and $p = 4$ pieces scale as ρ^{-1} and $\rho^{-(d+4)/d}$, respectively. Thus

$$\rho^{1+4/d} I_1^{\mathrm{odd}}(\rho) \tag{135}$$

contains the leading $\rho^{4/d}\phi(x)$ term from $p = 0$ and a finite local term from $p = 4$.

Even-dimensional case. For even d, the off-diagonal terms cancel because

$$L^{(p,q)}_{d-2+p-r} = L^{(p,q)}_r, \qquad \mathcal{I}^{(p,q,2)}_{d-2+p-r} = -\mathcal{I}^{(p,q,2)}_r. \tag{136}$$

Only the diagonal term $r^{(p)}_\star = (d-2+p)/2$ remains. Its coefficient can be written without any dimension-dependent summation as

$$L^{(p,q)}_{r_\star} = \sum_{m=0}^{p} \frac{(-1)^{\frac{d-2+p+2m}{2}} \Gamma(d-1) K^{(p,q)}_m}{\Gamma\left(\frac{d+p-2m}{2}\right)\Gamma\left(\frac{d-p+2m}{2}\right)}. \tag{137}$$

Applying the density operator to the logarithmic sector therefore gives

$$\widehat{\mathcal{E}}_d \mathcal{I}^{(p,q)}_{r_\star} \sim \frac{2L^{(p,q)}_{r_\star}}{d^2} \Gamma\left(q + \frac{d+p}{d}\right) Q'_d\left(-\frac{d+p}{d}\right) \sigma_d^{-(d+p)/d}, \tag{138}$$

and hence the complete even-dimensional near-region contribution is

$$I_1^{\rm even}(\rho) \sim \frac{\Omega_{d-2}\Gamma(d-1)}{2^{(d-4)/2}d^2} \sum_{p\in\{0,4\}} \sum_{q=0}^{2} (-1)^{(d-2+p)/2}\Gamma\left(q+\frac{d+p}{d}\right) Q_d'\left(-\frac{d+p}{d}\right) \sigma_d^{-(d+p)/d} \sum_{m=0}^{p} \frac{(-1)^m K_m^{(p,q)}}{\Gamma\left(\frac{d+p-2m}{2}\right)\Gamma\left(\frac{d-p+2m}{2}\right)}. \tag{139}$$

Thus, for even d, the $p=0$ logarithmic sector is converted into a ρ^{-1} term and gives the leading $\rho^{4/d}\phi(x)$ contribution, while the $p=4$ logarithmic sector is converted into a $\rho^{-(d+4)/d}$ term and gives a finite local contribution after multiplication by $\rho^{1+4/d}$.

Intermediate Basis and Covariant Reduction. For the retained sectors $p=0,4$, the coefficients $K_m^{(p,q)}$ are obtained by angular averaging the local expansion of the integrand. We perform this calculation in Cadabra using the adapted coordinates

$$x^\mu = te^\mu + r\widehat{n}^\mu, \qquad t = -\frac{u+v}{\sqrt{2}}, \qquad r = \frac{v-u}{\sqrt{2}}, \tag{140}$$

where e^μ is the unit timelike vector of the adapted frame and $\widehat{n}^\mu$ is a unit spatial vector orthogonal to it. The angular integration is performed using the angular-moment identities in Eq. (A49). Odd moments vanish, while even moments reduce to sums of products of the spatial projector $h^{\mu\nu} = \eta^{\mu\nu} + e^\mu e^\nu$. After angular averaging, the resulting curvature contractions are expressed in the adapted frame. In particular, contractions of a tensor $T_{\mu\nu\rho\sigma}$ with e^μ produce components such as $T_{0\nu\rho\sigma}$, where the index 0 denotes projection along the adapted timelike direction. Applying these reductions to the curvature tensors, their covariant derivatives, and the Taylor expansion of ϕ gives the following 30 intermediate structures. The first nine are manifest spacetime scalars, whereas $\mathcal{K}_{10},\dots,\mathcal{K}_{30}$ contain components defined relative to the adapted frame.

$$\begin{aligned}
&\mathcal{K}_1 = \phi && \mathcal{K}_2 = \Box^2\phi, && \mathcal{K}_3 = \Box R\phi, && \mathcal{K}_4 = R\Box\phi, && \mathcal{K}_5 = R_{\mu\nu}\nabla^\mu\nabla^\nu\phi,\\
&\mathcal{K}_6 = \nabla_\mu R\nabla^\mu\phi, && \mathcal{K}_7 = R^2\phi, && \mathcal{K}_8 = R_{\mu\nu}R^{\mu\nu}\phi, && \mathcal{K}_9 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\phi, && \mathcal{K}_{10} = \nabla_0\nabla_0\Box\phi,\\
&\mathcal{K}_{11} = \nabla_0\nabla_0 R\phi, && \mathcal{K}_{12} = R\nabla_0\nabla_0\phi, && \mathcal{K}_{13} = R_{\mu0}\nabla_0\nabla^\mu\phi, && \mathcal{K}_{14} = \nabla_0 R_{\mu0}\nabla^\mu\phi, && \mathcal{K}_{15} = \nabla_0 R\nabla_0\phi,\\
&\mathcal{K}_{16} = R_{\mu0}R^\mu{}_0\phi, && \mathcal{K}_{17} = \Box R_{00}\phi, && \mathcal{K}_{18} = R_{00}\Box\phi, && \mathcal{K}_{19} = \nabla_\mu R_{00}\nabla^\mu\phi, && \mathcal{K}_{20} = R_{\mu0\rho0}\nabla^\mu\nabla^\rho\phi,\\
&\mathcal{K}_{21} = \nabla^\mu R_{\mu0\nu0}\nabla^\nu\phi && \mathcal{K}_{22} = RR_{00}\phi, && \mathcal{K}_{23} = R_{\mu\nu\rho0}R^{\mu\nu\rho}{}_0\phi, && \mathcal{K}_{24} = R_{\mu\nu}R^\mu{}_0{}^\nu{}_0\phi, && \mathcal{K}_{25} = \nabla_0\nabla_0\nabla_0\nabla_0\nabla_0\phi,\\
&\mathcal{K}_{26} = \nabla_0\nabla_0 R_{00}\phi, && \mathcal{K}_{27} = R_{00}\nabla_0\nabla_0\nabla_0\phi, && \mathcal{K}_{28} = \nabla_0 R_{00}\nabla_0\phi, && \mathcal{K}_{29} = R_{00}^2\phi, && \mathcal{K}_{30} = R_{\mu0\rho0}R^\mu{}_0{}^\rho{}_0\phi.
\end{aligned}$$

The Cadabra result can then be written as

$$K_m^{(p,q)} = \sum_{A=1}^{30} c_{mA}^{(p,q)}\mathcal{K}_A. \tag{141}$$

We substitute this decomposition into the odd- and even-dimensional expressions for $I_1(\rho)$ and perform the complete weighted sums over p,q,m. Collecting the result by the intermediate basis, the Cadabra calculation shows that the net coefficients of $\mathcal{K}_{10},\dots,\mathcal{K}_{30}$ vanish identically in both parity cases. This cancellation is verified by the explicit coefficient summation and is consistent with the Lorentz invariance required of the final continuum operator. Consequently, only the scalar structures $\mathcal{K}_1,\dots,\mathcal{K}_9$ remain. Using the corresponding Gamma-function identities, we obtain the following expressions for the near-region integral.

$$\begin{aligned}
I_1^{\rm odd}(\rho) \sim \frac{\Omega_{d-2}\Gamma(d-1)}{2^{(d-2)/2}d}\Bigg[&\sigma_d^{-1}\frac{(-\frac{d}{2})\Gamma(-\frac{d}{2})}{\Gamma(\frac{d}{2})}\frac{3(-1)^{\frac{d-1}{2}}\Gamma(\frac{d}{2})}{2\sqrt{\pi}(d+2)\Gamma(\frac{d+5}{2})}\mathcal{K}_1 - \sigma_d^{-\frac{d+4}{d}}\frac{\Gamma(\frac{4}{d})\Gamma(-\frac{d}{2})}{\Gamma(\frac{d}{2})}\frac{(-1)^{\frac{d-1}{2}}\frac{d}{2}\Gamma(\frac{d}{2})}{\sqrt{\pi}\Gamma(\frac{d+5}{2})}\\
&\left(-\frac{1}{d(d+2)}\mathcal{K}_2 + \frac{2}{5}\frac{1}{d(d+2)}\mathcal{K}_3 + \frac{1}{d(d+1)}\mathcal{K}_4 + \frac{1}{d(d+1)}\mathcal{K}_5 + \frac{1}{d(d+2)}\mathcal{K}_6\right.\\
&\left.-\frac{1}{12}\frac{3d^2+12d+10}{d(d+1)^2(d+2)}\mathcal{K}_7 - \frac{1}{12}\frac{3d+4}{d(d+1)^2}\mathcal{K}_8 + \frac{1}{12}\frac{1}{d(d+1)}\mathcal{K}_9\right)\Bigg],
\end{aligned} \tag{142}$$

$$I_1^{\rm even}(\rho) \sim \frac{\Omega_{d-2}\Gamma(d-1)}{2^{(d-4)/2}d^2}\Bigg[\sigma_d^{-1}\frac{1}{\Gamma(\frac{d}{2})\Gamma(\frac{d}{2})}\frac{16}{(d+2)^2(d+4)}\mathcal{K}_1 - \sigma_d^{-\frac{d+4}{d}}\frac{\Gamma(\frac{4}{d})}{\Gamma(\frac{d}{2})\Gamma(\frac{d}{2})}\frac{2d}{(d+2)(d+4)}$$
$$\left(\frac{2}{d^2(d+2)}\mathcal{K}_2 - \frac{4}{5}\frac{1}{d^2(d+2)}\mathcal{K}_3 - \frac{2}{d^2(d+1)}\mathcal{K}_4 - \frac{2}{d^2(d+1)}\mathcal{K}_5 - \frac{2}{d^2(d+2)}\mathcal{K}_6\right.$$
$$\left.+\frac{1}{6}\frac{3d^2+12d+10}{d^2(d+1)^2(d+2)}\mathcal{K}_7 + \frac{1}{6}\frac{3d+4}{d^2(d+1)^2}\mathcal{K}_8 - \frac{1}{6}\frac{1}{d^2(d+1)}\mathcal{K}_9\right)\Bigg]. \tag{143}$$

h. Determination of α_d and β_d. Combining the Poisson-averaged expectation value in Eq. (6) with the near-region contribution in Eq. (65), we obtain

$$\left\langle B_4^{(d)}\phi(x)\right\rangle = \alpha_d\rho^{4/d}\phi(x) + \beta_d\rho^{1+4/d}I_1(\rho). \tag{144}$$

We use

$$\kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1}d(d-1)}, \qquad \sigma_d = \kappa_d\rho, \tag{145}$$

which implies

$$\rho^{1+4/d}\sigma_d^{-1} = \kappa_d^{-1}\rho^{4/d}, \qquad \rho^{1+4/d}\sigma_d^{-(d+4)/d} = \kappa_d^{-(d+4)/d}. \tag{146}$$

The normalization coefficients are fixed by requiring the continuum limit to contain $\Box^2\phi$ with unit coefficient and no divergent contribution proportional to $\phi(x)$. The coefficient α_d is chosen to cancel the leading density-dependent term, which scales as $\rho^{4/d}\phi(x)$, while β_d is fixed by requiring the coefficient of $\Box^2\phi$ to be unity. With these conditions imposed, the remaining finite terms determine the curvature-dependent contributions to the continuum operator. To express these normalization conditions explicitly, let $F_{1,d}^{\rm odd/even}$ denote the coefficient of $\sigma_d^{-1}\mathcal{K}_1$ and $F_{2,d}^{\rm odd/even}$ the coefficient of $\sigma_d^{-(d+4)/d}\mathcal{K}_2$ in the corresponding expressions for $I_1(\rho)$. The normalization conditions then read

$$\alpha_d + \beta_d\kappa_d^{-1}F_{1,d}^{\rm odd/even} = 0, \qquad \beta_d\kappa_d^{-(d+4)/d}F_{2,d}^{\rm odd/even} = 1. \tag{147}$$

The ratios of these coefficients, extracted from the odd- and even-dimensional asymptotic expressions Eqs. (142) and (143), respectively, are

$$\frac{F_{1,d}^{\rm odd}}{F_{2,d}^{\rm odd}} = -\frac{3d}{2\Gamma(4/d)}, \qquad \frac{F_{1,d}^{\rm even}}{F_{2,d}^{\rm even}} = -\frac{4d}{\Gamma(4/d)}. \tag{148}$$

Consequently, the normalization coefficients are

$$\alpha_d = \begin{cases} \frac{3d}{2\Gamma(4/d)}\kappa_d^{4/d}, & d \text{ odd}, \\ \frac{4d}{\Gamma(4/d)}\kappa_d^{4/d}, & d \text{ even}, \end{cases} \tag{149}$$

and

$$\beta_d = \begin{cases} \frac{(-1)^{(d-1)/2}2^{d/2}d\sqrt{\pi}(d+2)\Gamma\left(\frac{d+5}{2}\right)}{\Omega_{d-2}\Gamma(d-1)\Gamma(4/d)\Gamma(-d/2)}\kappa_d^{(d+4)/d}, & d \text{ odd}, \\ -\frac{2^{d/2-4}d^3(d+2)^2(d+4)\Gamma\left(\frac{d}{2}\right)^2}{\Omega_{d-2}\Gamma(d-1)\Gamma(4/d)}\kappa_d^{(d+4)/d}, & d \text{ even}. \end{cases} \tag{150}$$

E. Summary

For an integer spacetime dimension $d \geq 2$, the fourth-order causal-set operator is defined by the finite-layer ansatz

$$\mathcal{B}_4^{(d)}\phi(x) = \frac{1}{\ell^4}\left[\alpha_d\phi(x) + \beta_d\sum_{i=1}^{n_d}C_i^{(d)}\sum_{y\in L_i(x)}\phi(y)\right]. \tag{151}$$

The number of layers is

$$q_d = \left\lfloor\frac{d}{2}\right\rfloor + 2, \qquad n_d = \left\lfloor\frac{d}{2}\right\rfloor + 4. \tag{152}$$

The layer coefficients are generated by the polynomial

$$\mathcal{P}_d(z) = \sum_{i=1}^{n_d}C_i^{(d)}\frac{z^{i-1}}{(i-1)!}, \quad \widehat{\mathcal{E}}_d e^{-z} = \mathcal{P}_d(z)e^{-z}, \quad z = \rho V. \tag{153}$$

Here

$$\widehat{\mathcal{E}}_d = \mathcal{N}_d\left(D_\rho + \frac{d+2}{d}\right)\prod_{j=1}^{q_d}\left(D_\rho + \frac{2j}{d}\right), \qquad D_\rho = \rho\frac{\partial}{\partial\rho}, \tag{154}$$

with

$$\mathcal{N}_d = \frac{d^{q_d+1}}{2^{q_d}q_d!(d+2)}. \tag{155}$$

For a direct computation of the coefficients $C_i^{(d)}$, define

$$A_r^{(d)} = \frac{1}{q_d!}\begin{bmatrix} q_d+1 \\ r+1 \end{bmatrix}\left(\frac{d}{2}\right)^r, \qquad A_{-1}^{(d)} = A_{q_d+1}^{(d)} = 0. \tag{156}$$

The layer coefficients are then

$$C_i^{(d)} = (-1)^{i-1}(i-1)!\sum_{r=i-1}^{q_d+1}\left[A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right]\begin{Bmatrix} r \\ i-1 \end{Bmatrix}. \tag{157}$$

The geometric factor entering the normalization is

$$\Omega_{d-2} = \frac{2\pi^{(d-1)/2}}{\Gamma\left(\frac{d-1}{2}\right)}, \qquad \kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1}d(d-1)}. \tag{158}$$

The normalization coefficients are

$$\alpha_d = \begin{cases} \frac{3d}{2\Gamma(4/d)}\kappa_d^{4/d}, & d \text{ odd}, \\ \frac{4d}{\Gamma(4/d)}\kappa_d^{4/d}, & d \text{ even}, \end{cases} \tag{159}$$

and

$$\beta_d = \begin{cases} \frac{(-1)^{(d-1)/2}2^{d/2}d\sqrt{\pi}(d+2)\Gamma\left(\frac{d+5}{2}\right)}{\Omega_{d-2}\Gamma(d-1)\Gamma(4/d)\Gamma(-d/2)}\kappa_d^{(d+4)/d}, & d \text{ odd}, \\ -\frac{2^{d/2-4}d^3(d+2)^2(d+4)\Gamma\left(\frac{d}{2}\right)^2}{\Omega_{d-2}\Gamma(d-1)\Gamma(4/d)}\kappa_d^{(d+4)/d}, & d \text{ even}. \end{cases} \tag{160}$$

The continuum limit is

$$\begin{aligned} \lim_{\rho\to\infty}\left\langle \mathcal{B}_4^{(d)}\phi(x)\right\rangle = \Box^2\phi(x) &- \frac{2}{5}\Box R\phi \\ &- \frac{d+2}{d+1}R\Box\phi \\ &- \frac{d+2}{d+1}R_{\mu\nu}\nabla^\mu\nabla^\nu\phi \\ &- \nabla_\mu R\nabla^\mu\phi \\ &+ \frac{3d^2+12d+10}{12(d+1)^2}R^2\phi \\ &+ \frac{(3d+4)(d+2)}{12(d+1)^2}R_{\mu\nu}R^{\mu\nu}\phi \\ &- \frac{d+2}{12(d+1)}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\phi. \end{aligned} \tag{161}$$

III. FLUCTUATIONS

The continuum-limit calculation above concerns the expectation value of the causal-set operator. On an individual Poisson sprinkling, however, $B_4^{(d)}\phi(x)$ is a random variable. A finite continuum expectation value therefore does not by itself imply that a typical causal-set realization approaches the continuum operator. The relevant quantity is the variance

$$\mathrm{Var}\left(B_4^{(d)}\phi(x)\right) = \left\langle\left(B_4^{(d)}\phi(x)\right)^2\right\rangle - \left\langle B_4^{(d)}\phi(x)\right\rangle^2. \tag{162}$$

For the sharp operator,

$$B_4^{(d)}\phi(x) = \ell^{-4}\left[\alpha_d\phi(x) + \beta_d\sum_{i=1}^{n_d}C_i^{(d)}\sum_{y\in L_i(x)}\phi(y)\right], \tag{163}$$

the $\alpha_d\phi(x)$ term is deterministic and therefore does not contribute to the variance. The fluctuating part is a Poisson sum multiplied by ℓ^{-4}. Campbell-Mecke scaling gives

$$\mathrm{Var}\left(B_4^{(d)}\phi(x)\right) = \ell^{-8}\mathcal{V}_d(\phi,g) + o(\ell^{-8}), \tag{164}$$

where $\mathcal{V}_d(\phi,g)$ is dimensionless after the local variables have been rescaled by the sprinkling density. Indeed, the Poisson measure supplies one factor of ρ, while the region controlling the ultraviolet asymptotics has volume of order ρ^{-1}; hence the remaining integral is dimensionless and finite for a smooth compactly supported field and a finite-layer kernel. The characteristic scaling is therefore

$$\sqrt{\mathrm{Var}\left(B_4^{(d)}\phi(x)\right)} \sim \ell^{-4}. \tag{165}$$

This is the higher-order analogue of the fluctuation problem encountered for the causal-set d'Alembertian. The cancellations which render the expectation value finite act on the mean and do not provide a corresponding suppression of the shot noise. Thus increasing the sprinkling density alone does not produce a vanishing fluctuation amplitude for the sharp fourth-order operator.

A natural way to introduce statistical averaging is to separate the fundamental discreteness scale ℓ from a larger mesoscopic nonlocality scale ℓ_k,

$$\ell \ll \ell_k \ll L_{\mathrm{macro}}, \tag{166}$$

and to define

$$\rho_k := \ell_k^{-d}, \qquad \epsilon := \frac{\rho_k}{\rho} = \left(\frac{\ell}{\ell_k}\right)^d, \qquad 0<\epsilon<1. \tag{167}$$

The sharp operator is recovered at $\epsilon = 1$. For the fourth-order operator, the analogue of the Poisson-thinning construction is obtained by requiring the generating transform

$$F_\epsilon^{(4)}(\xi) := e^{-\xi}\sum_{n=0}^{\infty}f_d^{(4)}(n,\epsilon)\frac{\xi^n}{n!} = \epsilon^{1+4/d}e^{-\epsilon\xi}P_d(\epsilon\xi). \tag{168}$$

A convenient realization is

$$f_d^{(4)}(n,\epsilon) = \epsilon^{1+4/d}(1-\epsilon)^n\sum_{i=1}^{n_d}C_i^{(d)}\binom{n}{i-1}\left(\frac{\epsilon}{1-\epsilon}\right)^{i-1}. \tag{169}$$

Indeed, for a fixed layer i, putting $r = i-1$ gives

$$e^{-\xi}\sum_{n=r}^{\infty}(1-\epsilon)^n\binom{n}{r}\left(\frac{\epsilon}{1-\epsilon}\right)^r\frac{\xi^n}{n!} = e^{-\epsilon\xi}\frac{(\epsilon\xi)^r}{r!}, \tag{170}$$

and summing over i yields Eq. (168). The corresponding smeared operator is defined by

$$B_{\epsilon,4}^{(d)}\phi(x) = \rho_k^{4/d}\left[\alpha_d\phi(x) + \beta_d\sum_{y\prec x}f_d^{(4)}\left(n(x,y),\epsilon\right)\phi(y)\right]. \tag{171}$$

Its expectation value is

$$\left\langle B^{(d)}_{\epsilon,4}\phi(x)\right\rangle = \rho_k^{4/d}\left[\alpha_d\phi(x) + \beta_d\rho\int_{J^-(x)} d^dy\sqrt{-g(y)}\phi(y)F^{(4)}_\epsilon\left(\rho V(x,y)\right)\right]. \tag{172}$$

Therefore,

$$\left\langle B^{(d)}_{\epsilon,4}\phi(x)\right\rangle = \left\langle B^{(d)}_{4}\phi(x)\right\rangle\Big|_{\rho\to\rho_k}. \tag{173}$$

The smearing consequently does not modify the continuum operator extracted from the expectation value; it replaces the microscopic density by the effective density ρ_k.

The fluctuation problem is therefore reduced to determining how the variance scales with the two densities. The operator carries the prefactor $\rho_k^{4/d} = \ell_k^{-4}$, while the smeared kernel averages over a region containing, parametrically,

$$N_k \sim \rho\ell_k^d = \left(\frac{\ell_k}{\ell}\right)^d = \epsilon^{-1} \tag{174}$$

fundamental sprinkling elements. The resulting Poisson suppression gives

$$\mathrm{Var}\left(B^{(d)}_{\epsilon,4}\phi(x)\right) \sim \epsilon\ell_k^{-8} = \frac{\ell^d}{\ell_k^{d+8}}, \tag{175}$$

provided the dimensionless rescaled second-moment and connected two-point kernels remain bounded in the local scaling regime. Consequently,

$$\sqrt{\mathrm{Var}\left(B^{(d)}_{\epsilon,4}\phi(x)\right)} \sim \sqrt{\epsilon}\ell_k^{-4} = \frac{\ell^{d/2}}{\ell_k^{d/2+4}}. \tag{176}$$

Thus, the mesoscopic scale can suppress the sharp fluctuations. There are, however, two independent requirements in a simultaneous continuum limit:

$$\ell_k \longrightarrow 0, \qquad \frac{\ell}{\ell_k} \longrightarrow 0, \tag{177}$$

so that the operator remains local on macroscopic scales while averaging over many microscopic sprinkling elements, together with the stronger condition

$$L^8_{\mathrm{macro}}\frac{\ell^d}{\ell_k^{d+8}} \longrightarrow 0. \tag{178}$$

A sufficient hierarchy is therefore

$$\ell \ll \ell_k \ll L_{\mathrm{macro}}, \qquad L^8_{\mathrm{macro}}\frac{\ell^d}{\ell_k^{d+8}} \longrightarrow 0. \tag{179}$$

Hence the fourth-order causal-set operator admits a two-scale continuum limit in which its expectation value approaches the local continuum operator while its fluctuations are simultaneously driven to zero.

IV. CONCLUSION AND OUTLOOK

The results establish a direct finite-layer construction of a fourth-order causal-set operator whose Poisson-averaged continuum limit is a local covariant differential operator in arbitrary spacetime dimension $d \geq 2$. The central result is Eq. (161): a unit-normalized $\Box^2$ operator accompanied by a definite set of curvature couplings. These include terms involving $R\Box\phi$, $R_{\mu\nu}\nabla^\mu\nabla^\nu\phi$, $\nabla_\mu R\nabla^\mu\phi$, and $\Box R\phi$, together with the three quadratic curvature invariants multiplying ϕ. Their coefficients are explicit functions of the spacetime dimension and are fixed by the finite-layer prescription and the continuum normalization. Thus, the construction determines not only the leading fourth-derivative term, but also its complete curvature correction at the order considered. The covariance of this result is a substantive part of the calculation. The intermediate expansion contains tensor structures defined relative to an adapted frame, which cannot be interpreted individually as contributions to a local covariant operator. After the complete angular averaging and weighted coefficient summation, the coefficients of $\mathcal{K}_{10},\ldots,\mathcal{K}_{30}$ vanish identically in both even and odd dimensions. Only the nine covariant scalar structures remain. The resulting curvature completion is therefore fixed by the full causal-set kernel, rather than by a choice of frame or by matching selected terms to a prescribed continuum expression.

This construction complements earlier approaches in which higher-curvature information is obtained by successive applications of a lower-order causal-set d'Alembertian [6, 7, 17]. Iteration provides a useful route to particular curvature combinations. The present construction instead specifies a single finite-layer fourth-order kernel, with its own normalization and dimension-dependent curvature coefficients. It consequently provides an independent way to investigate which local four-derivative structures are selected by a causal-set prescription, without requiring the operator to be defined as a composition of lower-order operators. The geometric calculation also yields an Alexandrov-interval volume expansion in Riemann normal coordinates through fourth coordinate order, retaining non-vacuum curvature contributions in general dimension. Derivatives of curvature appear in intermediate expressions but do not survive in the complete volume integral at the order required here. This cancellation depends on treating the invariant volume element and the deformation of the causal boundary together. The resulting volume expansion supplies the geometric input for the continuum calculation and may also be useful in other higher-order analyses of causal-set

observables.

The constant-field specialization isolates the geometric density

$$\begin{aligned}\mathcal{Q}_4^{(d)}[g] := & -\frac{2}{5}\Box R + \frac{3d^2+12d+10}{12(d+1)^2}R^2 \\ & + \frac{(3d+4)(d+2)}{12(d+1)^2}R_{\mu\nu}R^{\mu\nu} \\ & - \frac{d+2}{12(d+1)}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}. \end{aligned} \tag{180}$$

Here g denotes the spacetime metric $g_{\mu\nu}$, and $\mathcal{Q}_4^{(d)}[g]$ denotes the scalar constructed from its curvature tensors. For a constant scalar field, the continuum operator therefore reduces to $\phi\mathcal{Q}_4^{(d)}[g]$. This is a definite local geometric contribution selected by the kernel in arbitrary dimension. It is a natural candidate for the bulk curvature sector of a corresponding causal-set action, but the local operator limit derived here does not, by itself, establish the complete continuum limit of that action. In particular, the treatment of total derivatives and the derivation of the associated boundary and corner contributions require a separate analysis.

The fluctuation analysis exposes a second issue that is not resolved by convergence of the expectation value. For the sharp operator, the variance scales as ℓ^{-8} and the root-mean-square fluctuation as ℓ^{-4}. The cancellations that yield a finite continuum mean do not, by themselves, suppress the fluctuations of individual Poisson sprinklings. The mesoscopic prescription addresses this distinction by introducing an averaging scale ℓ_k larger than the fundamental discreteness scale ℓ. Its expectation value is the sharp-operator expectation with the density replaced by $\rho_k = \ell_k^{-d}$, so it retains the same local continuum limit. Under the boundedness assumptions on the rescaled moment and connected two-point kernels, its variance scales as

$$\mathrm{Var}\left(B_{\epsilon,4}^{(d)}\phi(x)\right) \sim \frac{\ell^d}{\ell_k^{d+8}}. \tag{181}$$

Consequently, the continuum limit of the mean and the suppression of fluctuations can coexist in a suitable two-scale limit. With L_{macro} a fixed macroscopic length, the relevant hierarchy can be expressed as

$$\frac{\ell_k}{L_{\mathrm{macro}}} \longrightarrow 0, \qquad \frac{\ell}{\ell_k} \longrightarrow 0, \qquad L_{\mathrm{macro}}^8 \frac{\ell^d}{\ell_k^{d+8}} \longrightarrow 0. \tag{182}$$

The last condition is understood together with the boundedness assumptions stated in Sec. III. The fluctuation result is therefore conditional, but it identifies the scaling regime in which mesoscopic averaging can retain locality while suppressing the statistical noise.

Finally, two complementary routes to higher derivative order merit investigation. The first is to generalize the density-polynomial construction developed here, with suitable higher-degree Euler polynomials and finite-layer structures chosen to isolate higher powers of the d'Alembertian and determine their curvature completions directly. The second is to explore recursive compositions of the present fourth-order operator, following the strategy employed for the second-order causal-set d'Alembertian by de Brito, Eichhorn, and Pfeiffer [17]. In particular, repeated applications of $\mathcal{B}_4^{(d)}$, as well as mixed compositions of $\mathcal{B}_4^{(d)}$ with the second-order Benincasa–Dowker operator, could provide operators of nominally eighth and sixth derivative order, respectively. Both possible orderings of the mixed composition should be examined, since successive applications of the discrete operators need not yield identical curvature corrections. Comparing these recursive constructions with direct higher-order finite-layer ansätze may help identify the curvature invariants generated by causal order and the relations among their coefficients. Their continuum limits, normalization, and statistical fluctuations would need to be investigated independently. Such a programme could provide a systematic family of causal-set operators for exploring higher-derivative contributions in gravitational effective field theories.

The principal outcome is thus twofold: causal order determines a definite general-dimensional curvature completion for a direct fourth-order operator, and mesoscopic averaging provides a conditional route to controlling its fluctuations beyond the continuum limit of its expectation value.

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Appendix A: Alexandrov Volume

For two causally related events $y \prec x$, the Alexandrov interval is the intersection of the causal future of the earlier event with the causal past of the later event,

$$A(x,y) := J^+(y) \cap J^-(x). \tag{A1}$$

Its spacetime volume is

$$V(x,y) = \int_{A(x,y)} d^d z \sqrt{-g(z)}. \tag{A2}$$

Thus the Alexandrov volume depends on both the invariant volume element and the causal boundaries determined by the metric.

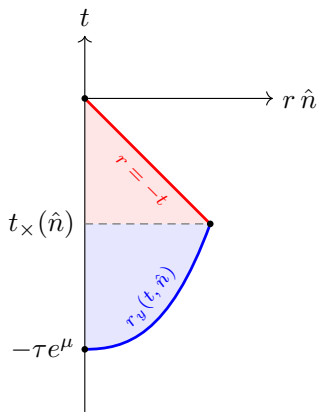


FIG. 2. Schematic Alexandrov interval in the RNC frame adapted to the timelike separation vector y^μ. The past null boundary of the RNC origin is exactly radial, as follows from the radial-geodesic property in Eq. (A4). In contrast, the future null boundary of y is generally deformed by curvature and is determined perturbatively using Synge's world function. The spatial radial direction $\hat{n}$ ranges over the complete $(d-2)$-sphere, and $t_\times(\hat{n})$ separates the two integration regions.

In the present calculation, the later endpoint is placed at the Riemann-normal-coordinate origin, $x^\mu = 0$. Although the separation vector y^μ is not generally aligned with the time coordinate of the original RNC system, it is timelike. We therefore introduce an orthonormal frame adapted to y^μ before determining the integration domain. In this frame the same timelike vector has purely temporal components, which places the endpoint at the centre of the spatial radial coordinates and permits the angular integration to be performed over the full sphere. The invariant proper-time separation is

$$\tau^2 := -\eta_{\mu\nu} y^\mu y^\nu = -2\sigma(0,y) > 0, \tag{A3}$$

where σ is Synge's world function [27, 28]. The relation between $\sigma(0,y)$ and the RNC coordinate vector y^μ follows from the radial-geodesic property in Eq. (A4). All curvature tensors appearing below are evaluated at the RNC origin.

1. Riemann Normal Coordinates

Riemann normal coordinates (RNC) based at the origin $x^\mu = 0$ are defined through the exponential map. Within the corresponding normal neighbourhood, a geodesic emanating from the origin with initial tangent v^μ is represented by the coordinate ray

$$x^\mu(\lambda) = \lambda v^\mu, \qquad x^\mu(0) = 0, \qquad \left.\frac{dx^\mu}{d\lambda}\right|_{\lambda=0} = v^\mu, \tag{A4}$$

where λ is an affine parameter. This straight-line property applies to geodesics through the RNC origin; a geodesic joining two generic points need not be a straight coordinate line [28]. At the origin, the metric and its first derivatives satisfy

$$g_{\mu\nu}(0) = \eta_{\mu\nu}, \qquad \Gamma^\mu{}_{\alpha\beta}(0) = 0, \qquad \partial_\rho g_{\mu\nu}(0) = 0. \tag{A5}$$

Consequently, the metric begins at quadratic order in the coordinates. Since the Alexandrov volume is required through total coordinate order $d+4$, the metric is retained through fourth coordinate order. The standard RNC metric expansion through this order is given in Refs. [29, 30]:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} - \frac{1}{3}R_{\mu\alpha\nu\beta}(0)x^\alpha x^\beta - \frac{1}{6}\nabla_\gamma R_{\mu\alpha\nu\beta}(0)x^\alpha x^\beta x^\gamma + \frac{1}{180}\left[8R_{\mu\alpha\beta\lambda}(0)R^\lambda{}_{\gamma\delta\nu}(0) - 9\nabla_\gamma\nabla_\delta R_{\mu\alpha\nu\beta}(0)\right]x^\alpha x^\beta x^\gamma x^\delta + \mathcal{O}(x^5). \tag{A6}$$

The corresponding invariant volume density follows by expanding the square root of the determinant of the metric in Eq. (A6):

$$\sqrt{-g(x)} = 1 - \frac{1}{6}R_{\mu\nu}x^\mu x^\nu - \frac{1}{12}\nabla_\rho R_{\mu\nu}x^\mu x^\nu x^\rho - \left[\frac{1}{40}\nabla_\rho\nabla_\sigma R_{\mu\nu} + \frac{1}{180}R_{\mu\alpha\nu}{}^\lambda R_{\rho\lambda\sigma}{}^\alpha - \frac{1}{72}R_{\mu\nu}R_{\rho\sigma}\right]x^\mu x^\nu x^\rho x^\sigma + \mathcal{O}(x^5). \tag{A7}$$

To make the endpoint geometry compatible with spherical integration, we now choose an orthonormal frame adapted to the timelike separation vector. Define

$$e^\mu := -\frac{y^\mu}{\tau}, \qquad e^\mu e_\mu = -1, \qquad y^\mu = -\tau e^\mu. \tag{A8}$$

Choose $d-1$ spacelike frame vectors $E^{\mu}{}_{(i)}$ satisfying

$$e_\mu E^{\mu}{}_{(i)} = 0, \qquad E_{(i)} \cdot E_{(j)} = \delta_{ij}, \qquad i,j = 1,\dots,d-1. \tag{A9}$$

Together with

$$E^{\mu}{}_{(0)} := e^\mu, \tag{A10}$$

they form an orthonormal frame at the RNC origin,

$$\eta_{\mu\nu} E^{\mu}{}_{(A)} E^{\nu}{}_{(B)} = \eta_{AB}. \tag{A11}$$

The corresponding adapted RNC coordinates are related to the original RNC coordinates by the constant linear transformation

$$x^\mu = E^{\mu}{}_{(A)} X^{(A)}. \tag{A12}$$

Hence the coordinate volume element is unchanged. In this basis, the endpoint has components

$$y^{(A)} = (-\tau, 0, \dots, 0). \tag{A13}$$

This is a choice of basis, not a restriction on the original vector y^μ. The invariant statement remains $y^\mu = -\tau e^\mu$. The spatial radial direction is described by a unit vector n^i,

$$n^i n_i = 1, \tag{A14}$$

with covariant spacetime representative

$$\widehat{n}^\mu := E^{\mu}{}_{(i)} n^i, \qquad e_\mu \widehat{n}^\mu = 0, \qquad \widehat{n}_\mu \widehat{n}^\mu = 1. \tag{A15}$$

The integration point is therefore written as

$$x^\mu = t e^\mu + r \widehat{n}^\mu, \tag{A16}$$

with

$$d^d x = dt dr r^{d-2} d\Omega_{d-2}. \tag{A17}$$

The spatial projector associated with the adapted frame is

$$h^{\mu\nu} = \eta^{\mu\nu} + e^\mu e^\nu. \tag{A18}$$

These relations are the only properties of the adapted frame needed below. In particular, the angular integrations are carried out entirely in the Euclidean spatial subspace orthogonal to e^μ and are subsequently reconstructed covariantly.

2. Integration Domain

The causal boundary of the Alexandrov interval consists of the future light cone of y and the past light cone of the origin. By the radial-geodesic property of RNC stated in Eq. (A4), the null geodesics emanating from the origin are exact coordinate rays. Consequently, the past null boundary of the origin is exactly radial:

$$r_{\rm p}(t) = -t, \qquad t \le 0. \tag{A19}$$

We work in a sufficiently small convex normal neighbourhood, so that the geodesic connecting the two endpoints is unique. The nontrivial boundary is the future light cone of the endpoint y. It is convenient to determine it from Synge's world function. For nearby points,

$$2\sigma(x,y) = \int_0^1 ds g_{\mu\nu}(z(s)) \dot{z}^\mu(s) \dot{z}^\nu(s), \qquad \dot{z}^\mu := \frac{dz^\mu}{ds}, \tag{A20}$$

where $z^\mu(s)$ is the unique affinely parametrized geodesic joining the two points, with $z^\mu(0) = y^\mu$ and $z^\mu(1) = x^\mu$. Introducing

$$\Delta^\mu := x^\mu - y^\mu, \tag{A21}$$

we assign coordinate order $\deg x^\mu = \deg y^\mu = 1$; a parenthesized label (k) on a quantities, z, r, t, F and G, denote its homogeneous component of coordinate order k. Thus, the geodesic is expanded perturbatively as

$$z^\mu(s) = z^\mu_{(1)}(s) + z^\mu_{(3)}(s) + \mathcal{O}(\lambda^4), \tag{A22}$$

where

$$z^\mu_{(1)}(s) = y^\mu + s\Delta^\mu, \tag{A23}$$

$$z^\mu_{(3)}(s) = \frac{1}{3} R^\mu{}_{\alpha\beta\nu} \Delta^\alpha \Delta^\beta y^\nu s(s-1). \tag{A24}$$

The higher-order terms are not written explicitly, since only the terms above are required for the world-function expansion used below. The construction satisfies the endpoint conditions order by order.

The null boundary is characterized by

$$\sigma(x,y) = 0. \tag{A25}$$

Using the adapted spherical coordinates (A16),

$$x^\mu = t e^\mu + r\widehat{n}^\mu, \qquad \widehat{n}^\mu = E^{\mu}{}_{(i)} n^i, \qquad y^\mu = -\tau e^\mu, \tag{A26}$$

define

$$F(r;t,n,y) := 2\sigma\left(te + r\widehat{n}, y\right). \tag{A27}$$

Its local expansion takes the form

$$F = F^{(2)} + F^{(4)} + F^{(5)} + F^{(6)} + \mathcal{O}(\lambda^7), \tag{A28}$$

where λ counts the coordinate order. In the adapted frame, the leading piece is simply

$$F^{(2)} = r^2 - (t+\tau)^2. \tag{A29}$$

Thus the physical leading-order radial solution is

$$r^{(1)}(t) = t + \tau, \qquad -\tau \le t \le 0. \tag{A30}$$

The algebraic root $r^{(1)} = -(t+\tau)$ is not part of the spherical coordinate domain and is discarded. For sufficiently small timelike separation, the physical curved-space branch remains the perturbative continuation of $r^{(1)}$.

We denote this physical radial boundary by

$$r_y(t,n) = r^{(1)}(t)+\delta r^{(3)}(t,n)+\delta r^{(4)}(t,n)+\delta r^{(5)}(t,n)+\mathcal{O}(\lambda^6). \tag{A31}$$

Substitution into $F(r_y;t,n,y) = 0$ gives the universal perturbative root formulas

$$\delta r^{(3)} = -\left.\frac{F^{(4)}}{\partial_r F^{(2)}}\right|_{r=r^{(1)}}, \tag{A32}$$

$$\delta r^{(4)} = -\left.\frac{F^{(5)}}{\partial_r F^{(2)}}\right|_{r=r^{(1)}}, \tag{A33}$$

$$\delta r^{(5)} = -\left.\frac{F^{(6)} + (\partial_r F^{(4)})\delta r^{(3)} + \frac{1}{2}(\partial_r^2 F^{(2)})(\delta r^{(3)})^2}{\partial_r F^{(2)}}\right|_{r=r^{(1)}}. \tag{A34}$$

All quantities on the right-hand side are evaluated at the corresponding leading-order root.

The radial boundary $r_y(t,n)$ intersects the exact past null boundary of the origin at a single transition time. Define

$$t_\times(n) \quad \text{by} \quad r_y(t_\times,n) = -t_\times. \tag{A35}$$

Equivalently,

$$G(t,n) := F(-t;t,n,y) = 0. \tag{A36}$$

At leading order,

$$G^{(2)}(t) = -\tau^2 - 2\tau t, \tag{A37}$$

so that

$$t_\times^{(1)} = -\frac{\tau}{2}. \tag{A38}$$

We write the curved transition time as

$$t_\times(n) = -\frac{\tau}{2} + \delta t^{(3)}(n) + \delta t^{(4)}(n) + \delta t^{(5)}(n) + \mathcal{O}(\lambda^6). \tag{A39}$$

The corrections are obtained from the same perturbative root construction,

$$\delta t^{(3)} = -\left.\frac{G^{(4)}}{\partial_t G^{(2)}}\right|_{t=-\tau/2}, \tag{A40}$$

$$\delta t^{(4)} = -\left.\frac{G^{(5)}}{\partial_t G^{(2)}}\right|_{t=-\tau/2}, \tag{A41}$$

$$\delta t^{(5)} = -\left.\frac{G^{(6)} + (\partial_t G^{(4)})\delta t^{(3)} + \frac{1}{2}(\partial_t^2 G^{(2)})(\delta t^{(3)})^2}{\partial_t G^{(2)}}\right|_{t=-\tau/2}. \tag{A42}$$

The resulting integration domain is therefore

$$-\tau \le t \le t_\times(n), \qquad 0 \le r \le r_y(t,n), \tag{A43}$$

followed by

$$t_\times(n) \le t \le 0, \qquad 0 \le r \le -t. \tag{A44}$$

The lower radial limit is consequently always $r = 0$ in the adapted-frame calculation. Curvature enters through the deformation of the first upper boundary and through the angular dependence of the transition time.

3. Volume Integral

Using the RNC volume density and the integration domain above, the Alexandrov volume is

$$V(0,y) = \int d\Omega_{d-2}\Big[\int_{-\tau}^{t_\times(n)} dt \int_0^{r_y(t,n)} dr r^{d-2}\sqrt{-g(t,r\widehat{n})} + \int_{t_\times(n)}^{0} dt \int_0^{-t} dr r^{d-2}\sqrt{-g(t,r\widehat{n})}\Big]. \tag{A45}$$

At this stage, all angular dependence is carried by the spatial unit vector n^i through $\widehat{n}^\mu = E^\mu{}_{(i)}n^i$. It is therefore useful to introduce the spatial projector

$$h^{\mu\nu} = \eta^{\mu\nu} + e^\mu e^\nu, \qquad e_\mu \widehat{n}^\mu = 0, \qquad \widehat{n}_\mu \widehat{n}^\mu = 1. \tag{A46}$$

The angular averages can then be written directly in spacetime form:

$$\int d\Omega_{d-2}\widehat{n}^\mu\widehat{n}^\nu = \frac{\Omega_{d-2}}{d-1}h^{\mu\nu}, \tag{A47}$$

$$\int d\Omega_{d-2}\widehat{n}^\mu\widehat{n}^\nu\widehat{n}^\rho\widehat{n}^\sigma = \frac{\Omega_{d-2}}{(d-1)(d+1)}\Big(h^{\mu\nu}h^{\rho\sigma} + h^{\mu\rho}h^{\nu\sigma} + h^{\mu\sigma}h^{\nu\rho}\Big), \tag{A48}$$

$$\int d\Omega_{d-2}\widehat{n}^{\mu_1}\cdots\widehat{n}^{\mu_{2k+1}} = 0. \tag{A49}$$

Higher even moments are obtained analogously as sums over all pairwise contractions with $h^{\mu\nu}$. The radial in-

tegrals are expanded consistently in coordinate order, retaining all terms that contribute through total order $d+4$.

After the angular integrations, the result is covariantized using

$$e^\mu = -\frac{y^\mu}{\tau}, \qquad h^{\mu\nu} = \eta^{\mu\nu} + \frac{y^\mu y^\nu}{\tau^2}. \tag{A50}$$

All remaining frame-dependent contractions are thereby expressed in terms of the invariant separation τ and the endpoint vector y^μ. The flat-space contribution is

$$V^{(d)}(y) = \frac{\Omega_{d-2}}{2^{d-1}d(d-1)}\tau^d, \qquad \Omega_{d-2} = \frac{2\pi^{(d-1)/2}}{\Gamma\left(\frac{d-1}{2}\right)}. \tag{A51}$$

There is no contribution at order $d+1$ and $d+3$. The local volume expansion required for the present calculation is therefore

$$V(0,y) = V^{(d)}(y) + V^{(d+2)}(y) + V^{(d+4)}(y) + \mathcal{O}(\lambda^{d+5}). \tag{A52}$$

It is convenient to collect the result in the form

$$V(0,y) = \frac{\Omega_{d-2}}{2^{d-1}d(d-1)}\tau^d\left[1 + \mathcal{V}_{(2)} + \mathcal{V}_{(4)} + \mathcal{O}(\lambda^5)\right], \tag{A53}$$

where

$$\mathcal{V}_{(2)} = \frac{d}{24(d+1)}\left[R_{\mu\nu}y^\mu y^\nu - \frac{R}{d+2}\tau^2\right], \tag{A54}$$

$$\begin{aligned}
\mathcal{V}_{(4)} = {} & \frac{d}{1152(d+1)(d+3)(d+4)}R^2\tau^4 \\
& + \frac{d}{720(d+1)(d+3)(d+4)}R_{\mu\nu}R^{\mu\nu}\tau^4 \\
& - \frac{d}{960(d+1)(d+3)(d+4)}R_{\alpha\beta\gamma\kappa}R^{\alpha\gamma\beta\kappa}\tau^4 \\
& - \frac{d}{576(d+1)(d+3)}RR_{\mu\nu}y^\mu y^\nu\tau^2 \\
& + \frac{d(7d+26)}{1440(d+1)(d+2)(d+3)}R_{\mu\nu}R^\mu{}_\sigma y^\nu y^\sigma\tau^2 \\
& - \frac{d(11d+34)}{1440(d+1)(d+2)(d+3)}R_{\mu\nu}R^\mu{}_\sigma{}^\nu{}_\beta y^\beta y^\sigma\tau^2 \\
& + \frac{d}{480(d+1)(d+3)}R_{\alpha\beta\gamma\kappa}R^{\alpha\gamma\beta}{}_\sigma y^\kappa y^\sigma\tau^2 \\
& + \frac{d(d+2)}{1152(d+1)(d+3)}R_{\mu\nu}R_{\rho\sigma}y^\mu y^\nu y^\rho y^\sigma \\
& + \frac{d(7d+22)}{2880(d+1)(d+3)}R_{\alpha\beta\gamma\kappa}R^\alpha{}_\nu{}^\gamma{}_\sigma y^\beta y^\kappa y^\nu y^\sigma.
\end{aligned} \tag{A55}$$

All curvature tensors and their covariant derivatives appearing in Eqs. (A54)–(A55) are evaluated at the RNC origin. The resulting expression is written in terms of the invariant proper-time separation and covariant contractions with the endpoint vector y^μ. Although the fourth-order Riemann normal-coordinate expansion of the metric contains terms involving covariant derivatives of the curvature tensor, these terms cancel upon performing the complete Alexandrov-volume integral at the order considered. Consequently, the final expansion depends only on the curvature combinations displayed in Eqs. (A54) and (A55).

Appendix B: Auxiliary Identities

1. Covariantization of the Scalar Taylor Expansion

All scalar-field derivatives below are evaluated at the RNC origin x. At this point,

$$g_{\mu\nu}(x) = \eta_{\mu\nu}, \qquad \partial_\rho g_{\mu\nu}(x) = 0, \qquad \Gamma^\rho_{\mu\nu}(x) = 0 \tag{B1}$$

Moreover, along the radial geodesic generated by a tangent vector y^μ, the tangent is parallel transported. Hence

$$\frac{d^n}{ds^n}\phi\left(\exp_x(sy)\right)\Big|_{s=0} = y^{\mu_1}\cdots y^{\mu_n}\nabla_{(\mu_1}\cdots\nabla_{\mu_n)}\phi(x). \tag{B2}$$

Since the ordinary partial derivatives of a scalar commute, comparison with the RNC Taylor expansion gives

$$\partial_\mu\phi = \nabla_\mu\phi \tag{B3}$$

$$\partial_\mu\partial_\nu\phi = \nabla_{(\mu}\nabla_{\nu)}\phi = \nabla_\mu\nabla_\nu\phi \tag{B4}$$

$$\partial_\mu\partial_\nu\partial_\rho\phi = \nabla_{(\mu}\nabla_\nu\nabla_{\rho)}\phi \tag{B5}$$

$$\partial_\mu\partial_\nu\partial_\rho\partial_\sigma\phi = \nabla_{(\mu}\nabla_\nu\nabla_\rho\nabla_{\sigma)}\phi \tag{B6}$$

More generally, the unsymmetrized fourth-order covariant derivative evaluated at the RNC origin is given by

$$\begin{aligned}
\nabla_\mu\nabla_\nu\nabla_\rho\nabla_\sigma\phi = {} & \partial_\mu\partial_\nu\partial_\rho\partial_\sigma\phi \\
& + \frac{1}{3}\left(\nabla_\mu R^\lambda{}_{\rho\sigma\nu} + \nabla_\mu R^\lambda{}_{\sigma\rho\nu}\right)\nabla_\lambda\phi \\
& + \frac{1}{3}\left(R^\lambda{}_{\rho\sigma\nu} + R^\lambda{}_{\sigma\rho\nu}\right)\nabla_\lambda\nabla_\mu\phi \\
& + \frac{1}{3}\left(R^\lambda{}_{\rho\sigma\mu} + R^\lambda{}_{\sigma\rho\mu}\right)\nabla_\lambda\nabla_\nu\phi \\
& + \frac{1}{3}\left(R^\lambda{}_{\nu\sigma\mu} + R^\lambda{}_{\sigma\nu\mu}\right)\nabla_\lambda\nabla_\rho\phi \\
& + \frac{1}{3}\left(R^\lambda{}_{\nu\rho\mu} + R^\lambda{}_{\rho\nu\mu}\right)\nabla_\lambda\nabla_\sigma\phi.
\end{aligned} \tag{B7}$$

The fourth-order contraction requires the commutation of covariant derivatives. With

$$\Box\phi := g^{\mu\nu}\nabla_\mu\nabla_\nu\phi, \tag{B8}$$

one finds

$$g^{\mu\rho}g^{\nu\sigma}\nabla_\mu\nabla_\nu\nabla_\rho\nabla_\sigma\phi = \Box^2\phi + R^{\mu\nu}\nabla_\mu\nabla_\nu\phi + \frac{1}{2}\nabla^\mu R\nabla_\mu\phi \tag{B9}$$

The remaining non-equivalent contraction gives the same result, and therefore the fully symmetrized fourth-order contraction is

$$g^{\mu\nu}g^{\rho\sigma}\nabla_{(\mu}\nabla_\rho\nabla_\nu\nabla_{\sigma)}\phi = \Box^2\phi + \frac{2}{3}R^{\mu\nu}\nabla_\mu\nabla_\nu\phi + \frac{1}{3}\nabla^\mu R\nabla_\mu\phi \tag{B10}$$

2. Integral Identities

We use the Gamma function

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \tag{B11}$$

together with the digamma function

$$\psi(z) := \frac{d}{dz} \ln \Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}. \tag{B12}$$

For later use, we write the lower and upper incomplete Gamma functions as $\gamma(s,z)$ and $\Gamma(s,z)$, respectively, so that

$$\gamma(s,z) + \Gamma(s,z) = \Gamma(s). \tag{B13}$$

a. Exponential Integral. For $n > -1$,

$$\int_0^R x^n e^{-x} dx = \gamma(n+1, R) = \Gamma(n+1) - \Gamma(n+1, R). \tag{B14}$$

For integer $n \geq 0$, this may also be written as

$$\int_0^R x^n e^{-x} dx = n! \left[1 - e^{-R} \sum_{k=0}^{n} \frac{R^k}{k!} \right]. \tag{B15}$$

For fixed n and $R \to \infty$,

$$\Gamma(n+1, R) = R^n e^{-R} \left[1 + \frac{n}{R} + \mathcal{O}\left(R^{-2}\right) \right], \tag{B16}$$

and therefore

$$\int_0^R x^n e^{-x} dx = \Gamma(n+1) - R^n e^{-R} \left[1 + \frac{n}{R} + \mathcal{O}\left(R^{-2}\right) \right]. \tag{B17}$$

b. General-Dimensional Double Integral. Consider

$$\mathcal{I}^{(d)}_{m,n}(a,\sigma) := \int_0^a dv \int_0^v du u^m v^n \exp\left[-\sigma (uv)^{d/2} \right]. \tag{B18}$$

We introduce

$$t = uv, \qquad u = \frac{t}{v}, \qquad du = \frac{dt}{v}. \tag{B19}$$

For fixed v, the original bound $0 \leq u \leq v$ gives $0 \leq t \leq v^2$. Therefore,

$$\mathcal{I}^{(d)}_{m,n} = \int_0^a dv \int_0^{v^2} dt t^m v^{n-m-1} \exp\left[-\sigma t^{d/2} \right]. \tag{B20}$$

The transformed integration region is therefore $0 \leq v \leq a$ and $0 \leq t \leq v^2$. Reversing the order of integration, t runs from 0 to a^2, while for fixed t one has $\sqrt{t} \leq v \leq a$. Hence

$$\mathcal{I}^{(d)}_{m,n} = \int_0^{a^2} dt t^m e^{-\sigma t^{d/2}} \int_{\sqrt{t}}^a dv v^{n-m-1}. \tag{B21}$$

For $m \neq n$,

$$\int_{\sqrt{t}}^a dv v^{n-m-1} = \frac{a^{n-m} - t^{(n-m)/2}}{n-m}, \tag{B22}$$

and therefore

$$\mathcal{I}^{(d)}_{m,n} = \frac{1}{n-m} \left[a^{n-m} \int_0^{a^2} dt t^m e^{-\sigma t^{d/2}} - \int_0^{a^2} dt t^{(m+n)/2} e^{-\sigma t^{d/2}} \right]. \tag{B23}$$

Define

$$K_p(a,\sigma) := \int_0^{a^2} dt t^p e^{-\sigma t^{d/2}}. \tag{B24}$$

Using

$$z = \sigma t^{d/2}, \qquad t = \left(\frac{z}{\sigma}\right)^{2/d}, \qquad dt = \frac{2}{d} \sigma^{-2/d} z^{2/d-1} dz, \tag{B25}$$

we obtain

$$\begin{aligned} K_p(a,\sigma) &= \frac{2}{d} \sigma^{-2(p+1)/d} \int_0^{\sigma a^d} dz z^{2(p+1)/d - 1} e^{-z} \\ &= \frac{2}{d} \sigma^{-2(p+1)/d} \gamma\left(\frac{2(p+1)}{d}, \sigma a^d \right). \end{aligned} \tag{B26}$$

Consequently, for $m \neq n$,

$$\mathcal{I}^{(d)}_{m,n}(a,\sigma) = \frac{2}{d(n-m)} \left[a^{n-m} \sigma^{-2(m+1)/d} \gamma\left(\frac{2(m+1)}{d}, \sigma a^d \right) - \sigma^{-(m+n+2)/d} \gamma\left(\frac{m+n+2}{d}, \sigma a^d \right) \right]. \tag{B27}$$

For the diagonal case $m = n$, the integral (B21) becomes

$$\int_{\sqrt{t}}^a \frac{dv}{v} = \ln a - \frac{1}{2} \ln t \implies \mathcal{I}^{(d)}_{m,m} = \int_0^{a^2} dt t^m e^{-\sigma t^{d/2}} \left(\ln a - \frac{1}{2} \ln t \right). \tag{B28}$$

Using

$$K_m(a,\sigma) = \frac{2}{d} \sigma^{-s_m} \gamma(s_m, \sigma a^d), \qquad s_m := \frac{2(m+1)}{d}, \qquad \int_0^{a^2} dt t^m \ln t e^{-\sigma t^{d/2}} = \frac{\partial K_m}{\partial m}, \tag{B29}$$

we find

$$\mathcal{I}^{(d)}_{m,m}(a,\sigma) = \frac{2}{d^2}\sigma^{-2(m+1)/d}\left[\gamma\left(\frac{2(m+1)}{d}, \sigma a^d\right)\ln(\sigma a^d) - \frac{\partial}{\partial s}\gamma\left(s, \sigma a^d\right)\Big|_{s=2(m+1)/d}\right]. \tag{B30}$$

For the large-σa^d limit, writing $\gamma(s,\sigma a^d) = \Gamma(s) - \Gamma(s,\sigma a^d)$, give us

$$\mathcal{I}^{(d)}_{m,n}(a,\sigma) = \frac{2}{d(n-m)}\left[a^{n-m}\Gamma\left(\frac{2(m+1)}{d}\right)\sigma^{-2(m+1)/d} - \Gamma\left(\frac{m+n+2}{d}\right)\sigma^{-(m+n+2)/d}\right] + \mathcal{O}\left(e^{-\sigma a^d}\right), \qquad m \neq n. \tag{B31}$$

and

$$\mathcal{I}^{(d)}_{m,m}(a,\sigma) = \frac{2}{d^2}\sigma^{-2(m+1)/d}\Gamma\left(\frac{2(m+1)}{d}\right)\left[\ln(\sigma a^d) - \psi\left(\frac{2(m+1)}{d}\right)\right] + \mathcal{O}\left(e^{-\sigma a^d}\right), \qquad m = n \tag{B32}$$

where

$$\psi(s) := \frac{d}{ds}\ln\Gamma(s) = \frac{\Gamma'(s)}{\Gamma(s)} \tag{B33}$$

is the digamma function.

For later use, it is convenient to identify the two asymptotic sectors in Eq. (B31):

$$\mathcal{I}^{(1)}_{m,n} = \frac{2a^{n-m}}{d(n-m)}\Gamma\left(\frac{2(m+1)}{d}\right)\sigma^{-2(m+1)/d}, \tag{B34}$$

$$\mathcal{I}^{(2)}_{m,n} = -\frac{2}{d(n-m)}\Gamma\left(\frac{m+n+2}{d}\right)\sigma^{-(m+n+2)/d}. \tag{B35}$$

The second sector is antisymmetric under interchange of the two indices:

$$\mathcal{I}^{(2)}_{m,n} = -\mathcal{I}^{(2)}_{n,m}, \qquad m \neq n. \tag{B36}$$

Consequently, a symmetric coefficient multiplying the pair (m,n) and (n,m) gives an exact cancellation of the second asymptotic sector. For the diagonal case,

$$\mathcal{I}^{(d)}_{m,m} \sim \sigma^{-2(m+1)/d}\left[A_m \ln\sigma + B_m\right], \tag{B37}$$

which is the origin of the logarithmic sectors discussed in Sec. II.

3. Euler-Operator Identity

Let

$$H := x\frac{d}{dx} \tag{B38}$$

denote the Euler operator, and let

$$P(t) = \sum_{k=0}^{m} c_k t^k \tag{B39}$$

be a polynomial. For $x > 0$ and $a \in \mathbb{C}$,

$$H(x^a) = a x^a, \tag{B40}$$

while

$$H\left(x^a \ln x\right) = a x^a \ln x + x^a. \tag{B41}$$

It follows by induction that

$$H^k\left(x^a \ln x\right) = a^k x^a \ln x + k a^{k-1} x^a, \tag{B42}$$

with the second term understood to vanish for $k = 0$. Hence, by linearity,

$$P(H)\left(x^a \ln x\right) = P(a) x^a \ln x + P'(a) x^a, \tag{B43}$$

where $P'(a)$ denotes the ordinary derivative of $P(t)$ with respect to its argument. In particular, if λ is a double root of the Euler polynomial, then

$$(D_\rho - \lambda)^2\left[\rho^\lambda (A\ln\rho + B)\right] = 0. \tag{B44}$$

A simple root removes the logarithm but leaves a term proportional to ρ^λ; a double root removes both.

4. Endpoint Values of the Euler Polynomial

For use in the odd- and even-dimensional asymptotic expressions Eqs. (142) and (143), we evaluate the required values of the eigenvalue polynomial associated with the normalized Euler operator.

$$Q_d(\lambda) = N_d\left(\lambda + \frac{d+2}{d}\right)\prod_{j=1}^{q_d}\left(\lambda + \frac{2j}{d}\right) \tag{B45}$$

with

$$q_d = \left\lfloor\frac{d+2}{2}\right\rfloor + 1, \qquad N_d = \frac{d^{q_d+1}}{2^{q_d} q_d!(d+2)} \tag{B46}$$

For odd $d = 2s+1$, one has $q_d = s+2$, and therefore

$$\begin{aligned} Q_d(-1) &= N_d \frac{2}{d}\prod_{j=1}^{s+2}\frac{2j-(2s+1)}{d} \\ &= \frac{3(-1)^s\Gamma\left(\frac{d}{2}\right)}{2\sqrt{\pi}(d+2)\Gamma\left(\frac{d+5}{2}\right)} \end{aligned} \tag{B47}$$

and

$$Q_d\left(-\frac{d+4}{d}\right) = N_d\left(-\frac{2}{d}\right)\prod_{j=1}^{s+2}\frac{2j-(2s+5)}{d}$$
$$= -(-1)^s\frac{\Gamma\left(\frac{d+2}{2}\right)}{\sqrt{\pi}\Gamma\left(\frac{d+5}{2}\right)} \tag{B48}$$

For even $d = 2s$, one has $q_d = s+2$, and both relevant arguments are roots of Q_d,

$$Q_d(-1) = 0, \qquad Q_d\left(-\frac{d+4}{d}\right) = 0 \tag{B49}$$

Their derivatives are obtained by removing the corresponding vanishing factor:

$$Q'_d(-1) = N_d\frac{2}{d}\prod_{\substack{j=1\\ j\neq s}}^{s+2}\frac{j-s}{s}$$
$$= -\frac{16(-1)^{d/2}}{(d+2)^2(d+4)} \tag{B50}$$

and

$$Q'_d\left(-\frac{d+4}{d}\right) = N_d\left(-\frac{2}{d}\right)\prod_{j=1}^{s+1}\frac{j-s-2}{s}$$
$$= (-1)^{d/2}\frac{2d}{(d+2)(d+4)} \tag{B51}$$

These are the only values of Q_d and Q'_d required in the continuum-limit calculation.

5. Finite-Difference Inversion Formula

Let $F(x)$ be a polynomial of degree at most $n-1$. Introduce the discrete-binomial basis

$$e_i(x) := (-1)^{i-1}\binom{x}{i-1}, \qquad i = 1,\dots,n. \tag{B52}$$

Since $\{e_i(x)\}_{i=1}^n$ forms a basis for polynomials of degree at most $n-1$, we may write

$$F(x) = \sum_{i=1}^n C_i e_i(x). \tag{B53}$$

Define the forward-difference operator

$$\Delta f(x) := f(x+1) - f(x). \tag{B54}$$

Using

$$\Delta\binom{x}{k} = \binom{x}{k-1}, \tag{B55}$$

one obtains

$$\Delta^m e_i(0) = \begin{cases}(-1)^{i-1}, & m = i-1,\\ 0, & m \neq i-1.\end{cases} \tag{B56}$$

Applying Δ^{i-1} to Eq. (B53) and evaluating at $x = 0$ therefore isolates C_i:

$$C_i = (-1)^{i-1}\Delta^{i-1}F(0). \tag{B57}$$

The $(i-1)$ -th forward difference has the standard expansion

$$\Delta^{i-1}F(0) = \sum_{k=0}^{i-1}\binom{i-1}{k}(-1)^{i-1-k}F(k), \tag{B58}$$

and hence

$$C_i = \sum_{k=0}^{i-1}\binom{i-1}{k}(-1)^k F(k), \qquad i = 1,\dots,n. \tag{B59}$$

This inversion formula determines the coefficients of the discrete-binomial basis directly from the values of the polynomial at the integers $0,1,\dots,n-1$.

6. Finite Binomial-Sum Identity

For $a \notin \{0,1,\dots,n\}$, define

$$S_n(a) := \sum_{j=0}^n \frac{(-1)^j\binom{n}{j}}{a-j}. \tag{B60}$$

For $\mathrm{Re}(a) > n$, using

$$\frac{1}{a-j} = \int_0^1 t^{a-j-1}dt, \tag{B61}$$

and the binomial theorem,

$$S_n(a) = \int_0^1 dt t^{a-1}\sum_{j=0}^n(-1)^j\binom{n}{j}t^{-j} \tag{B62}$$
$$= (-1)^n\int_0^1 dt t^{a-n-1}(1-t)^n \tag{B63}$$
$$= (-1)^n B(a-n, n+1) \tag{B64}$$
$$= (-1)^n\frac{\Gamma(n+1)\Gamma(a-n)}{\Gamma(a+1)}. \tag{B65}$$

Both sides are meromorphic functions of a, and the result therefore extends by analytic continuation to

$$\sum_{j=0}^n\frac{(-1)^j\binom{n}{j}}{a-j} = (-1)^n\frac{\Gamma(n+1)\Gamma(a-n)}{\Gamma(a+1)} \tag{B66}$$

for $a \notin \{0,1,\dots,n\}$. In particular, no restriction on the relative magnitudes of a and n is required.

7. Closed-Form Layer Coefficients

We now derive the layer coefficients appearing in Eq. (5) directly from the Euler operator (102). Let

$$q := q_d. \tag{B67}$$

The normalized density operator is

$$\widehat{\mathcal{E}}_d = \frac{d^{q+1}}{2^q q!(d+2)} \left(D_\rho + \frac{d+2}{d}\right) \prod_{j=1}^{q} \left(D_\rho + \frac{2j}{d}\right). \tag{B68}$$

To expand this operator in powers of D_ρ, introduce the unsigned Stirling numbers of the first kind, $\left[{n \atop k}\right]$, defined by

$$x^{\overline{n}} = x(x+1)\cdots(x+n-1) = \sum_{k=0}^{n} \left[{n \atop k}\right] x^k, \tag{B69}$$

where $x^{\overline{n}}$ denotes the rising factorial. They satisfy the recurrence relation

$$\left[{n+1 \atop k}\right] = \left[{n \atop k-1}\right] + n\left[{n \atop k}\right]. \tag{B70}$$

Since

$$\prod_{j=1}^{q} \left(D_\rho + \frac{2j}{d}\right) = \left(\frac{2}{d}\right)^q \prod_{j=1}^{q} \left(\frac{d}{2}D_\rho + j\right), \tag{B71}$$

we may write

$$\begin{aligned} \prod_{j=1}^{q} \left(D_\rho + \frac{2j}{d}\right) &= \left(\frac{2}{d}\right)^q \left(\frac{d}{2}D_\rho + 1\right)^{\overline{q}} \\ &= \left(\frac{2}{d}\right)^q \frac{\left(\frac{d}{2}D_\rho\right)^{\overline{q+1}}}{\frac{d}{2}D_\rho} \\ &= \left(\frac{2}{d}\right)^q \sum_{r=0}^{q} \left[{q+1 \atop r+1}\right] \left(\frac{d}{2}\right)^r D_\rho^r. \end{aligned} \tag{B72}$$

Therefore, defining

$$A_r^{(d)} := \frac{1}{q!}\left[{q+1 \atop r+1}\right]\left(\frac{d}{2}\right)^r, \qquad A_{-1}^{(d)} = A_{q+1}^{(d)} = 0, \tag{B73}$$

the normalized Euler operator becomes

$$\widehat{\mathcal{E}}_d = \sum_{r=0}^{q+1} \left[A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right] D_\rho^r. \tag{B74}$$

We next determine the polynomial generated by this operator. Writing

$$z = \rho V, \tag{B75}$$

one has

$$D_\rho = z\frac{d}{dz}. \tag{B76}$$

The standard relation between powers of the Euler derivative and ordinary derivatives is

$$\left(z\frac{d}{dz}\right)^r = \sum_{m=0}^{r} \left\{{r \atop m}\right\} z^m \frac{d^m}{dz^m}, \tag{B77}$$

where $\left\{{n \atop k}\right\}$ are the Stirling numbers of the second kind.

The Stirling numbers of the second kind are defined by the expansion of the ordinary power in falling factorials,

$$x^n = \sum_{k=0}^{n} \left\{{n \atop k}\right\} x^{\underline{k}}, \tag{B78}$$

where

$$x^{\underline{k}} := x(x-1)\cdots(x-k+1). \tag{B79}$$

They obey the recurrence

$$\left\{{n+1 \atop k}\right\} = \left\{{n \atop k-1}\right\} + k\left\{{n \atop k}\right\}. \tag{B80}$$

Since

$$\frac{d^m}{dz^m}e^{-z} = (-1)^m e^{-z}, \tag{B81}$$

it follows that

$$D_\rho^r e^{-z} = e^{-z}\sum_{m=0}^{r}(-1)^m \left\{{r \atop m}\right\} z^m. \tag{B82}$$

Applying Eq. (B74) therefore yields

$$\widehat{\mathcal{E}}_d e^{-z} = e^{-z}\sum_{m=0}^{q+1}(-1)^m \left[\sum_{r=m}^{q+1}\left(A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right)\left\{{r \atop m}\right\}\right] z^m. \tag{B83}$$

Comparing this with

$$\mathcal{P}_d(z)e^{-z} = \sum_{i=1}^{q+2} C_i^{(d)} \frac{z^{i-1}}{(i-1)!}e^{-z}, \tag{B84}$$

and setting $m = i-1$, gives the closed-form expression

$$C_i^{(d)} = (-1)^{i-1}(i-1)! \sum_{r=i-1}^{q+1}\left[A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right]\left\{{r \atop i-1}\right\}, \tag{B85}$$

where $i = 1, \ldots, q+2$. Equivalently,

$$\begin{aligned} C_i^{(d)} = \frac{(-1)^{i-1}(i-1)!}{q!} \sum_{r=i-1}^{q+1} \Bigg[&\left[{q+1 \atop r+1}\right]\left(\frac{d}{2}\right)^r \\ &+ \frac{d}{d+2}\left[{q+1 \atop r}\right]\left(\frac{d}{2}\right)^{r-1}\Bigg]\left\{{r \atop i-1}\right\}. \end{aligned} \tag{B86}$$

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